

- 1 The heights x cm of 100 boys in Year 7 at a school are summarised in the table below.

Height	$125 \leq x \leq 140$	$140 < x \leq 145$	$145 < x \leq 150$	$150 < x \leq 160$	$160 < x \leq 170$
Frequency	25	29	24	18	4

- (i) Estimate the number of boys who have heights of at least 155 cm.

[2]

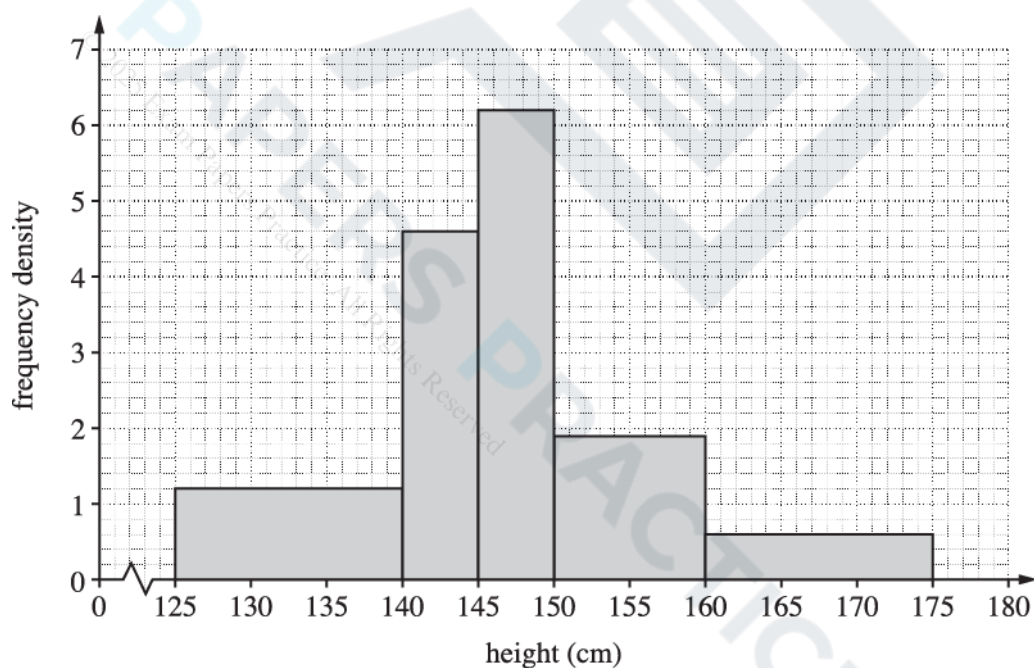
- (ii) Calculate an estimate of the median height of the 100 boys.

[3]

- (iii) Draw a histogram to illustrate the data.

[5]

The histogram below shows the heights of 100 girls in Year 7 at the same school.



- (iv) How many more girls than boys had heights exceeding 160 cm?

[3]

(v) Calculate an estimate of the mean height of the 100 girls.

[5]

2 The stem and leaf diagram illustrates the heights in metres of 25 young oak trees.

3		4	6	7	8	9	9		
4		0	2	2	3	4	6	8	9
5		0	1	3	5	8			
6		2	4	5					
7		4	6						
8		1							

Key: 4|2 represents 4.2

(i) State the type of skewness of the distribution.

[1]

(ii) Use your calculator to find the mean and standard deviation of these data.

[3]

(iii) Determine whether there are any outliers.

[4]

3 The birth weights in kilograms of 25 female babies are shown below, in ascending order.

1.39	2.50	2.68	2.76	2.82	2.82	2.84	3.03	3.06	3.16	3.16	3.24	3.32
3.36	3.40	3.54	3.56	3.56	3.70	3.72	3.72	3.84	4.02	4.24	4.34	

(i) Find the median and interquartile range of these data.

[3]

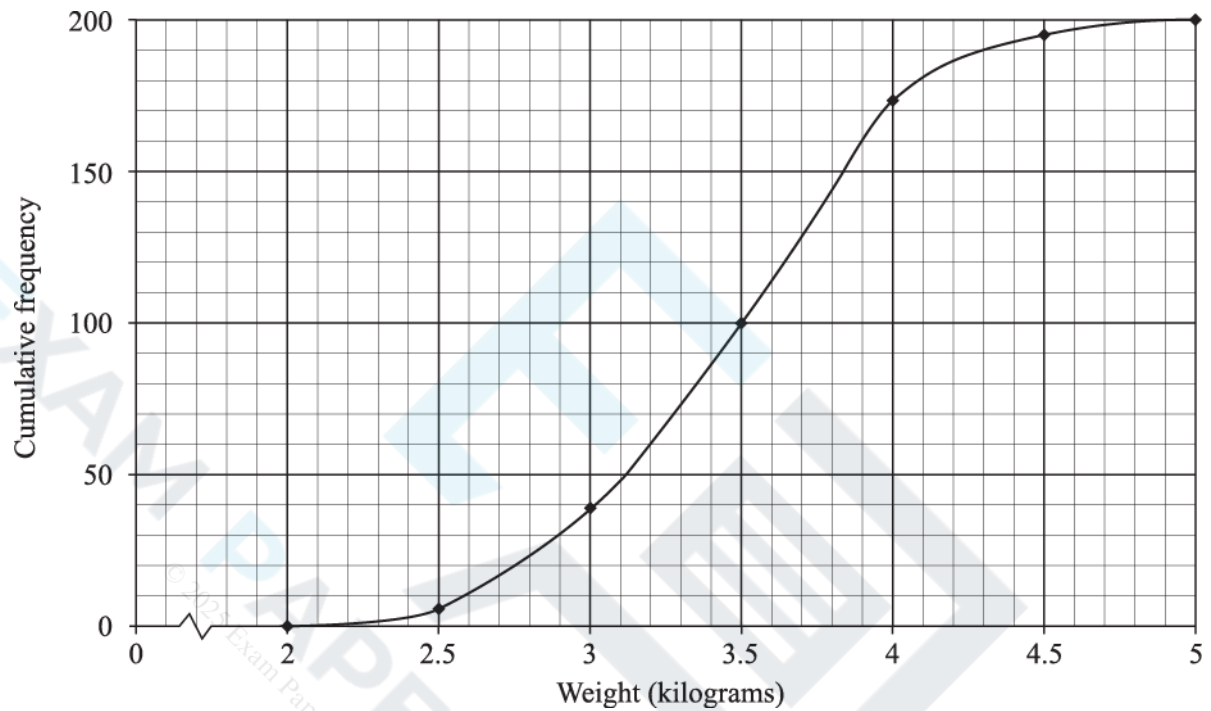
(ii) Draw a box and whisker plot to illustrate the data.

[3]

(iii) Show that there is exactly one outlier. Discuss whether this outlier should be removed from the data.

[4]

The cumulative frequency curve below illustrates the birth weights of 200 male babies.



(iv) Find the median and interquartile range of the birth weights of the male babies.

[3]

(v) Compare the weights of the female and male babies.

[2]

(vi) Two of these male babies are chosen at random. Calculate an estimate of the probability that both of these babies weigh more than any of the female babies.

[3]

- 4 The weights, w grams, of a random sample of 60 carrots of variety A are summarised in the table below.

Weight	$30 \leq w < 50$	$50 \leq w < 60$	$60 \leq w < 70$	$70 \leq w < 80$	$80 \leq w < 90$
Frequency	11	10	18	14	7

- (i) Draw a histogram to illustrate these data.

[5]

- (ii) Calculate estimates of the mean and standard deviation of w .

[4]

- (iii) Use your answers to part (ii) to investigate whether there are any outliers.

[3]

The weights, x grams, of a random sample of 50 carrots of variety B are summarised as follows.

$$n = 50$$

$$\sum x = 3624.5$$

$$\sum x^2 = 265\,416$$

- (iv) Calculate the mean and standard deviation of x .

[3]

- (v) Compare the central tendency and variation of the weights of varieties A and B.

[2]

- 5 The ages, x years, of the senior members of a running club are summarised in the table below.

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Age (x)	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$	$60 \leq x < 70$	$70 \leq x < 80$	$80 \leq x < 90$
Frequency	10	30	42	23	9	5	1

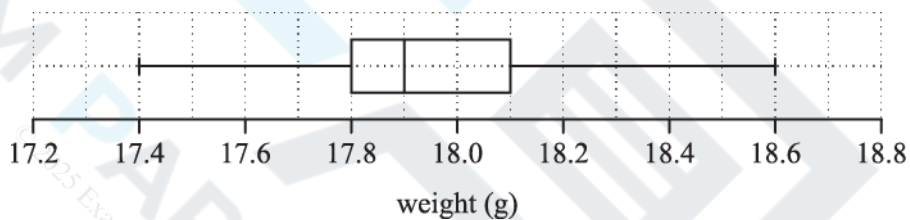
- (i) Draw a cumulative frequency diagram to illustrate the data.

[5]

- (ii) Use your diagram to estimate the median and interquartile range of the data.

[3]

- 6 The box and whisker plot below summarises the weights in grams of the 20 chocolates in a box.



- (i) Find the interquartile range of the data and hence determine whether there are any outliers at either end of the distribution.

[5]

Ben buys a box of these chocolates each weekend. The chocolates all look the same on the outside, but 7 of them have orange centres, 6 have cherry centres, 4 have coffee centres and 3 have lemon centres.

One weekend, each of Ben's 3 children eats one of the chocolates, chosen at random.

- (ii) Calculate the probabilities of the following events.

A all 3 chocolates have orange centres

B all 3 chocolates have the same centres

[6]

(iii) Find $P(A|B)$ and $P(B|A)$.

[3]

The following weekend, Ben buys an identical box of chocolates and again each of his 3 children eats one of the chocolates, chosen at random.

(iv) Find the probability that, on both weekends, the 3 chocolates that they eat all have orange centres.

[2]

(v) Ben likes all of the chocolates except those with cherry centres. On another weekend he is the first of his family to eat some of the chocolates. Find the probability that he has to select more than 2 chocolates before he finds one that he likes.

[3]

7 At a tourist information office the numbers of people seeking information each hour over the course of a 12-hour day are shown below.

6 25 38 39 31 18 35 31 33 15 21 28

(i) Construct a sorted stem and leaf diagram to represent these data.

[3]

(ii) State the type of skewness suggested by your stem and leaf diagram.

[1]

(iii) For these data find the median, the mean and the mode. Comment on the usefulness of the mode in this case.

[4]

8 An online store has a total of 930 different types of women's running shoe on sale. The prices in pounds of the types of women's running shoe are summarised in the table below.

Price (£ x)	$10 \leq x \leq 40$	$40 < x \leq 50$	$50 < x \leq 60$	$60 < x \leq 80$	$80 < x \leq 200$
Frequency	147	109	182	317	175

(i) Calculate estimates of the mean and standard deviation of the shoe prices.

[4]

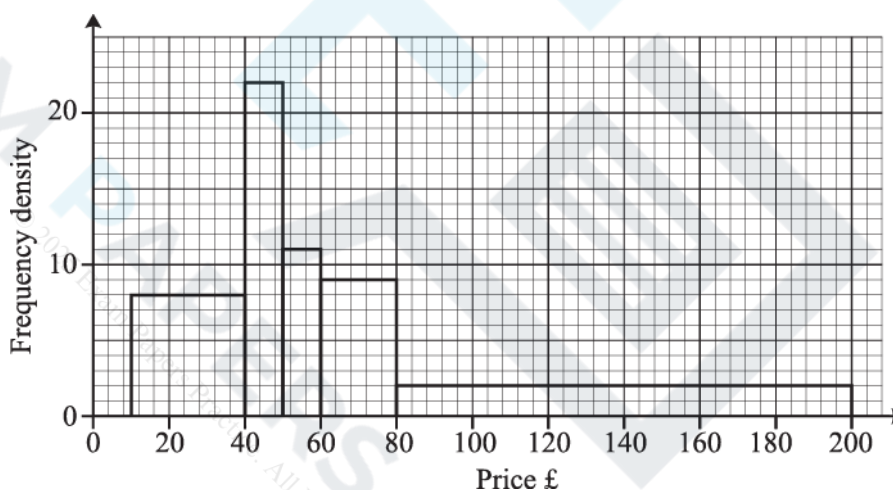
(ii) Calculate an estimate of the percentage of types of shoe that cost at least £100.

[3]

(iii) Draw a histogram to illustrate the data.

[5]

The corresponding histogram below shows the prices in pounds of the 990 types of men's running shoe on sale at the same online store.



(iv) State the type of skewness shown by the histogram for men's running shoes.

[1]

(v) Martin is investigating the percentage of types of shoe on sale at the store that cost more than £100. He believes that this percentage is greater for men's shoes than for women's shoes. Estimate the percentage for men's shoes and comment on whether you can be certain which percentage is higher.

[3]

(vi) You are given that the mean and standard deviation of the prices of men's running shoes are £68.83 and £42.93 respectively. Compare the central tendency and variation of the prices of men's and women's running shoes at the store.

[2]

- 9 The stem and leaf diagram illustrates the weights in grams of 20 house sparrows.

25	0		
26	0	5	8
27	7	9	
28	1	4	5
29	0	0	2
30	7	7	
31	6		
32	0	4	7
33	3	3	

Key: 27 | 7 represents 27.7 grams

- (i) Find the median and interquartile range of the data.

[3]

- (ii) Determine whether there are any outliers.

[4]

- 10 A researcher believes that there may be negative association between the quantity of fertiliser used and the percentage of the population who live in rural areas in different countries. The data below show the percentage of the population who live in rural areas and the fertiliser use measured in kg per hectare, for a random sample of 11 countries.

Percentage of population	33	6	58	35	81	69	61	7	74	71	17
Fertiliser use	76	44	6	68	3	10	7	176	5	137	157

- (i) Draw a scatter diagram to illustrate the data.

[3]

- (ii) Explain why it might not be valid to carry out a test based on the product moment correlation coefficient in this case.

[2]

- (iii) Calculate the value of Spearman's rank correlation coefficient.

[5]

- (iv) Carry out a hypothesis test at the 1% significance level to investigate the researcher's belief.

[6]

- (v) Explain the meaning of '1% significance level'.

[1]

- (vi) In order to carry out a test based on Spearman's rank correlation coefficient, what modelling assumptions, if any, are required about the underlying distribution?

[1]

- 11 Alison selects 10 of her male friends. For each one she measures the distance between his eyes. The distances, measured in mm, are as follows:

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51 57 58 59 61 64 64 65 67 68

The mean of these data is 61.4. The sample standard deviation is 5.232, correct to 3 decimal places.

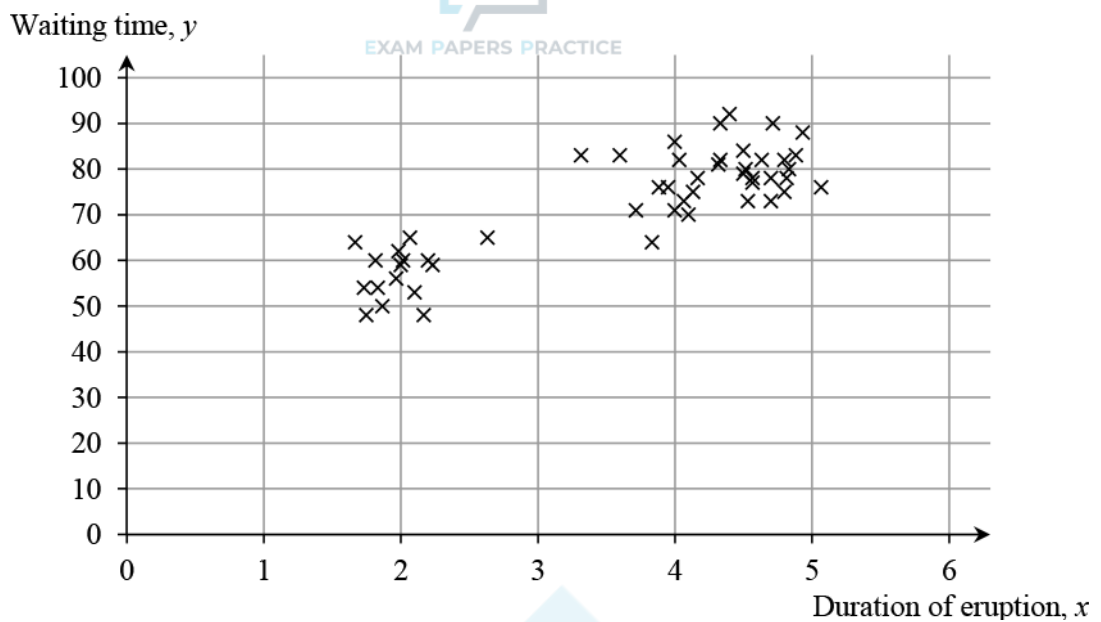
One of the friends decides he does not want his measurement to be used. Alison replaces his measurement with the measurement from another male friend. This increases the mean to 62.0 and reduces the standard deviation. Give a possible value for the measurement which has been removed and find the measurement which has replaced it.

[3]

- 12 A geyser is a hot spring which erupts from time to time. For two geysers, the duration of each eruption, x minutes, and the waiting time until the next eruption, y minutes, are recorded.
- (a) For a random sample of 50 eruptions of the first geyser, the correlation coefficient between x and y is 0.758. The critical value for a 2-tailed hypothesis test for correlation at the 5% level is 0.279. Explain whether or not there is evidence of correlation in the population of eruptions.

[2]

The scatter diagram in Fig. 9 shows the data from a random sample of 50 eruptions of the second geyser.



- (b) Stella claims the scatter diagram shows evidence of correlation between duration of eruption and waiting time. Make two comments about Stella's claim. [2]

- 13 The quality control department of a battery manufacturing company checks the lifetimes of the batteries produced by the company. The lifetimes, x minutes, for a random sample of 80 'Superstrength' batteries are shown in the table below.

Lifetime	$160 \leq x < 165$	$165 \leq x < 168$	$168 \leq x < 170$	$170 \leq x < 172$	$172 \leq x < 175$	$175 \leq x < 180$
Frequency	5	14	20	21	16	4

- (a) Estimate the proportion of these batteries which have a lifetime of at least 174.0 minutes. [2]

(b) Use the data in the table to estimate

- the sample mean,
- the sample standard deviation.

The data in the table on the previous page are represented in the following histogram:

Frequency density

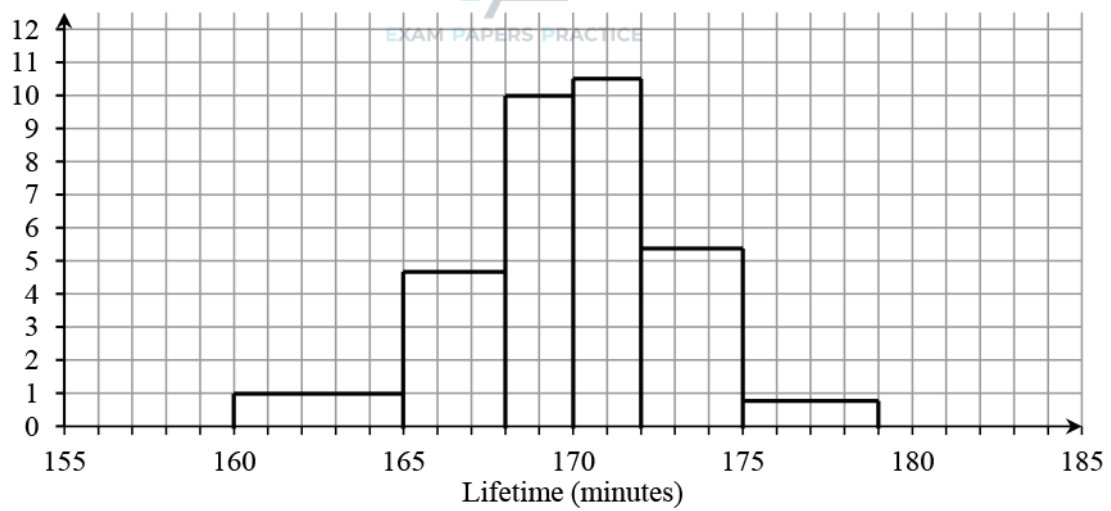


Fig. 15

A quality control manager models the data by a Normal distribution with the mean and standard deviation you calculated in part (b).

(c) Comment briefly on whether the histogram supports this choice of model.

[2]

(d) (i) Use this model to estimate the probability that a randomly selected battery will have a lifetime of more than 174.0 minutes.

(ii) Compare your answer with your answer to part (a).

[3]

The company also manufactures 'Ultrapower' batteries, which are stated to have a mean lifetime of 210 minutes.

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- (e) A random sample of 8 Ultrapower batteries is selected. The mean lifetime of these batteries is 207.3 minutes. Carry out a hypothesis test at the 5% level to investigate whether the mean lifetime is as high as stated. You should use the following hypotheses $H_0 : \mu = 210$, $H_1 : \mu < 210$, where μ represents the population mean for Ultrapower batteries. You should assume that the population is Normally distributed with standard deviation 3.4.

[5]



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14 Fig. 16.1, Fig. 16.2 and Fig. 16.3 show some data about life expectancy, including some from the pre-release data set (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>).

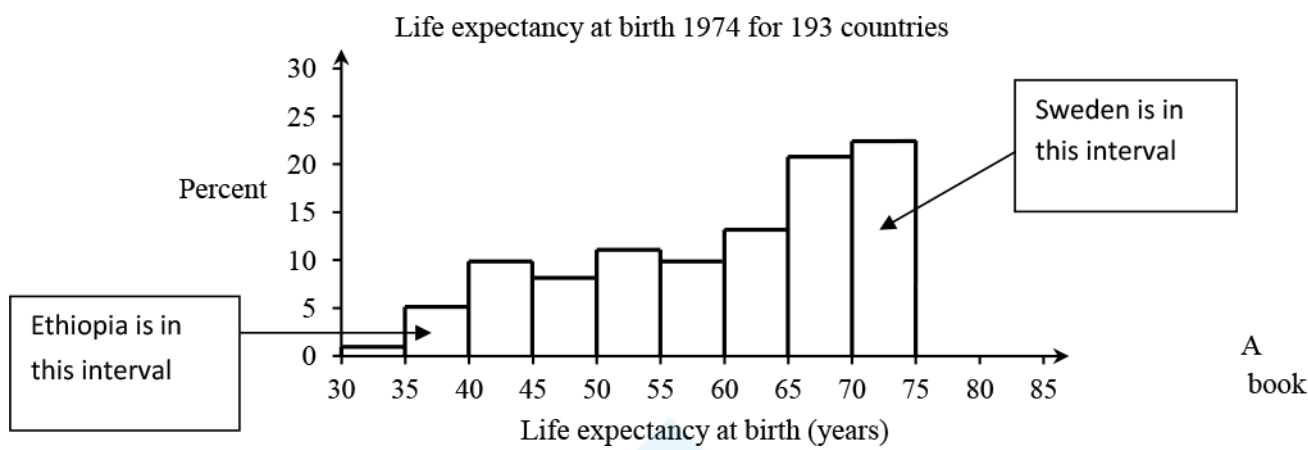


Fig. 16.1

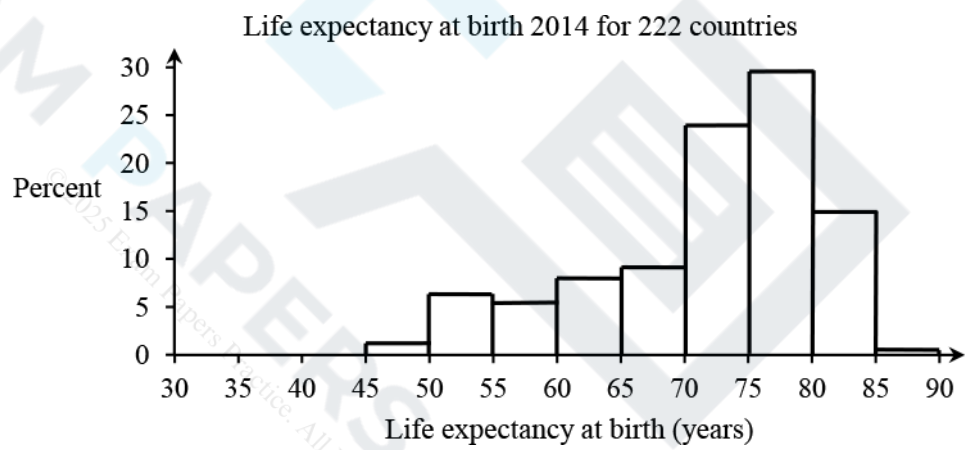
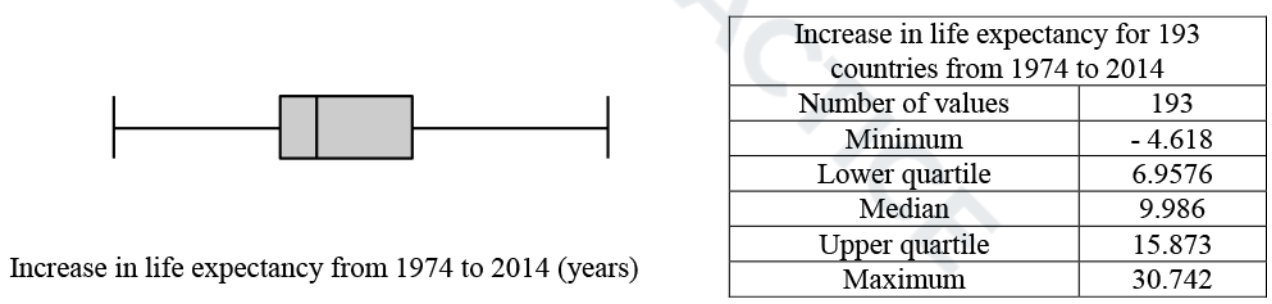


Fig. 16.2



Source: CIA World Factbook and Gapminder

- (a) Comment on the shapes of the distributions of life expectancy at birth in 2014 and 1974. [2]
- (b) (i) The minimum value shown in the box plot is negative. What does a negative value indicate? [1]
- (ii) What feature of Fig 16.3 suggests that a Normal distribution would **not** be an appropriate model for increase in life expectancy from one year to another year? [1]
- (iii) Software has been used to obtain the values in the table in Fig. 16.3. Decide whether the level of accuracy is appropriate. Justify your answer. [1]
- (iv) John claims that for half the people in the world their life expectancy has improved by 10 years or more. Explain why Fig. 16.3 does not provide conclusive evidence for John's claim. [1]
- (c) Decide whether the maximum increase in life expectancy from 1974 to 2014 is an outlier. Justify your answer. [3]

Here is some further information from the pre-release data set (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>).

Country	Life expectancy at birth in 2014
Ethiopia	60.8
Sweden	81.9

(d) (i) Estimate the change in life expectancy at birth for Ethiopia between 1974 and 2014.

(ii) Estimate the change in life expectancy at birth for Sweden between 1974 and 2014.

(iii) Give **one** possible reason why the answers to parts (i) and (ii) are so different.

[4]

Fig.16.4 shows the relationship between life expectancy at birth in 2014 and 1974.

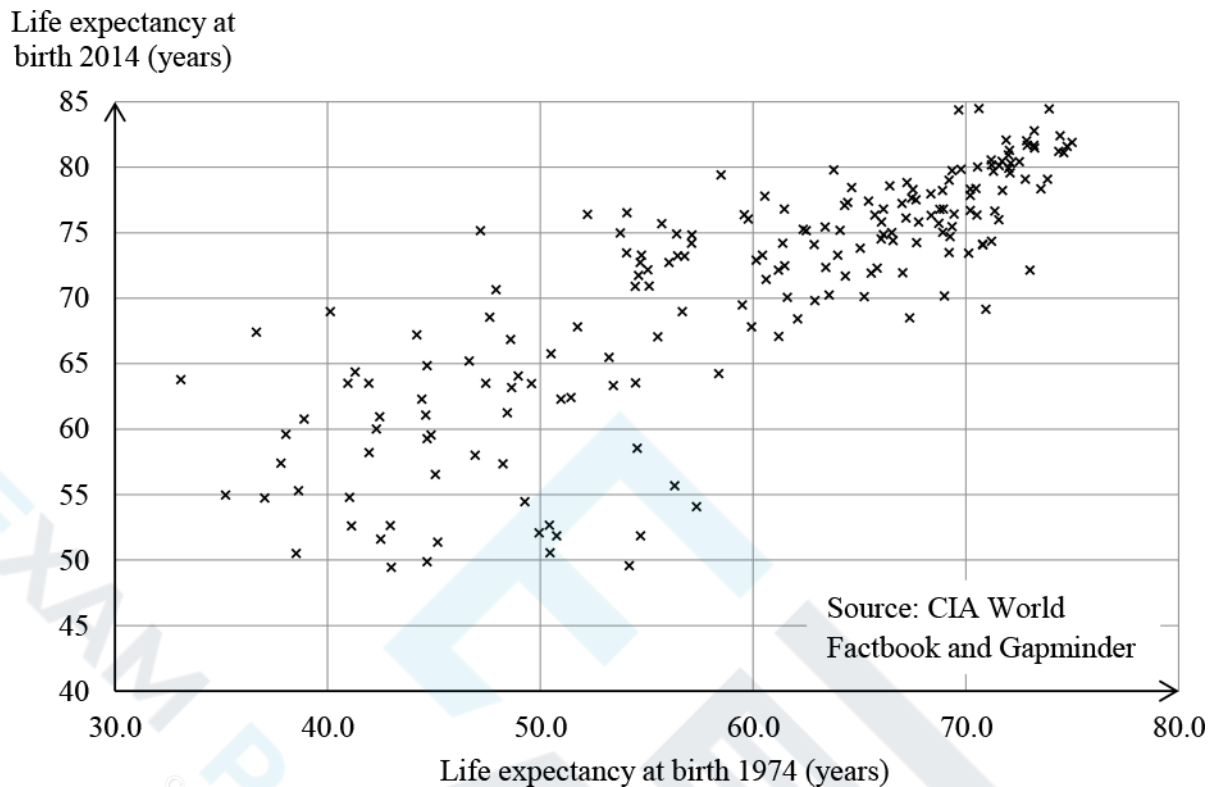


Fig. 16.4

A spreadsheet gives the following linear model for all the data in Fig 16.4.

$$(\text{Life expectancy at birth 2014}) = 30.98 + 0.67 \times (\text{Life expectancy at birth 1974})$$

The life expectancy at birth in 1974 for the region that now constitutes the country of South Sudan was 37.4 years. The value for this country in 2014 is not available.

(e) (i) Use the linear model to estimate the life expectancy at birth in 2014 for South Sudan.

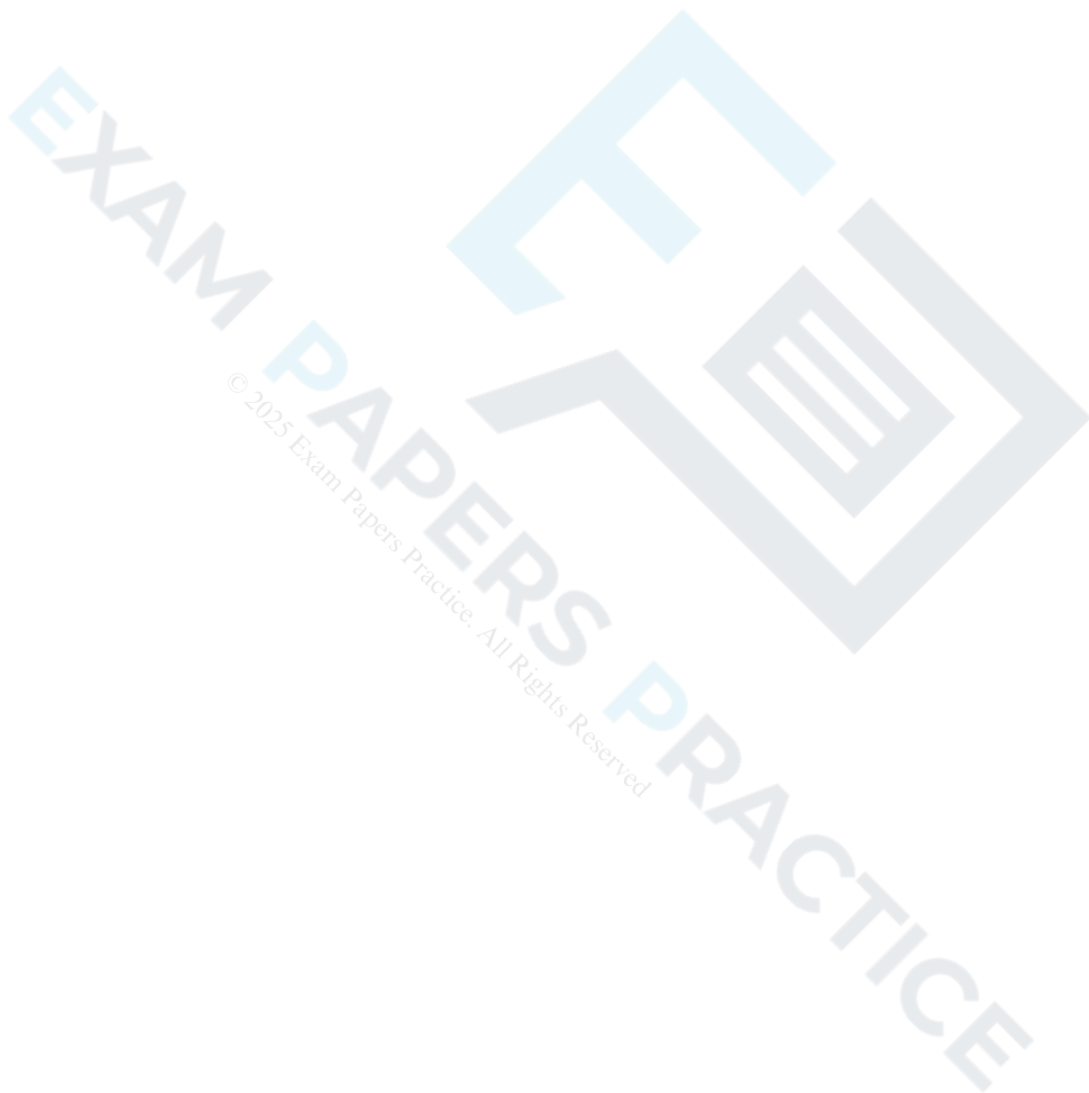
[2]

- (ii) Give two reasons why your answer to part (i) is not likely to be an accurate estimate for the life expectancy at birth in 2014 for South Sudan. You should refer to **both** information from Fig 16.4 and your knowledge of the large data set.

[2]

- (f) In how many of the countries represented in Fig. 16.4 did life expectancy drop between 1974 and 2014? Justify your answer.

[3]



- 15 A farmer has 200 apple trees. She is investigating the masses of the crops of apples from individual trees. She decides to select a sample of these trees and find the mass of the crop for each tree.

(a) Explain how she can select a random sample of 10 different trees from the 200 trees. [2]

The masses of the crops from the 10 trees, measured in kg, are recorded as follows.

23.5 27.4 26.2 29.0 25.1 27.4 26.2 28.3 38.1 24.9

(b) For these data find

- the mean,
- the sample standard deviation.

[2]

(c) Show that there is one outlier at the upper end of the data. How should the farmer decide whether to use this outlier in any further analysis of the data? [3]

- 16 Fig. 9.1 shows box and whisker diagrams which summarise the birth rates per 1000 people for all the countries in three of the regions as given in the pre-release data set. The diagrams were drawn as part of an investigation comparing birth rates in different regions of the world.

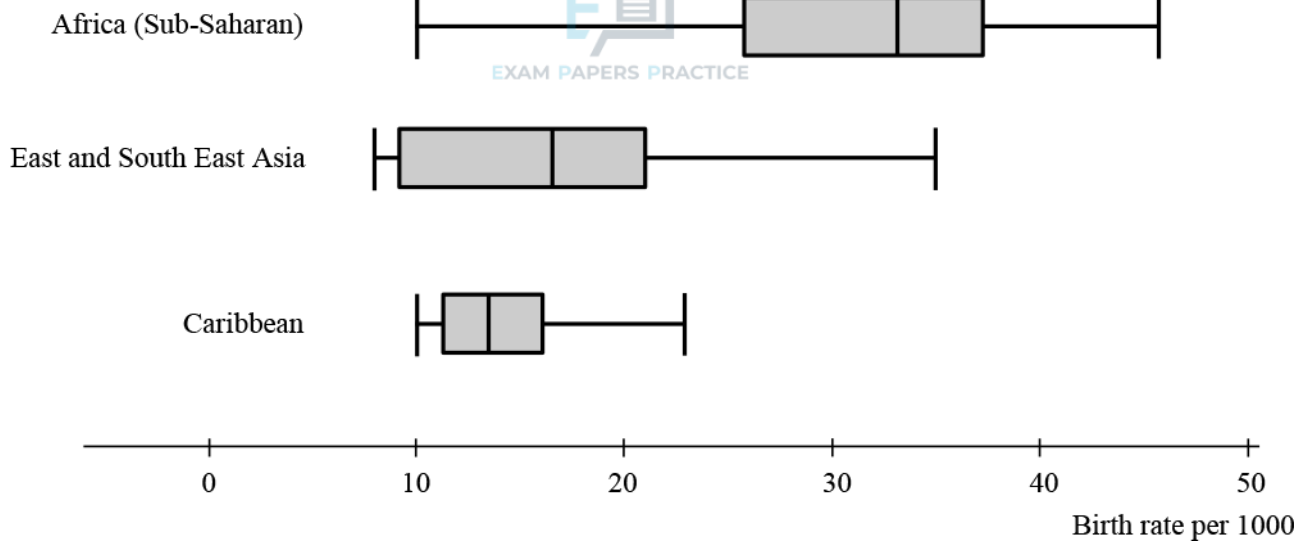


Fig. 9.1

- (a) Discuss the distributions of birth rates in these regions of the world. Make three different statements. You should refer to **both** information from the box and whisker diagrams **and** your knowledge of the large data set. To access the Large Data Set please go to <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls> [3]

- (b) The birth rates for all the countries in Australasia are shown below.

Country	Birth rate per 1000
Australia	12.19
New Zealand	13.4
Papua New Guinea	24.89

- (i) Explain why the calculation below is not a correct method for finding the birth rate per 1000 for Australasia as a whole.

$$\frac{12.19 + 13.4 + 24.89}{3} \approx 16.83$$

[1]

- (ii) Without doing any calculations, explain whether the birth rate per 1000 for Australasia as a whole is higher or lower than 16.83. [1]

The scatter diagram in Fig. 9.2 shows birth rate per 1000 and physicians/1000 population for all the countries in the pre-release data set.

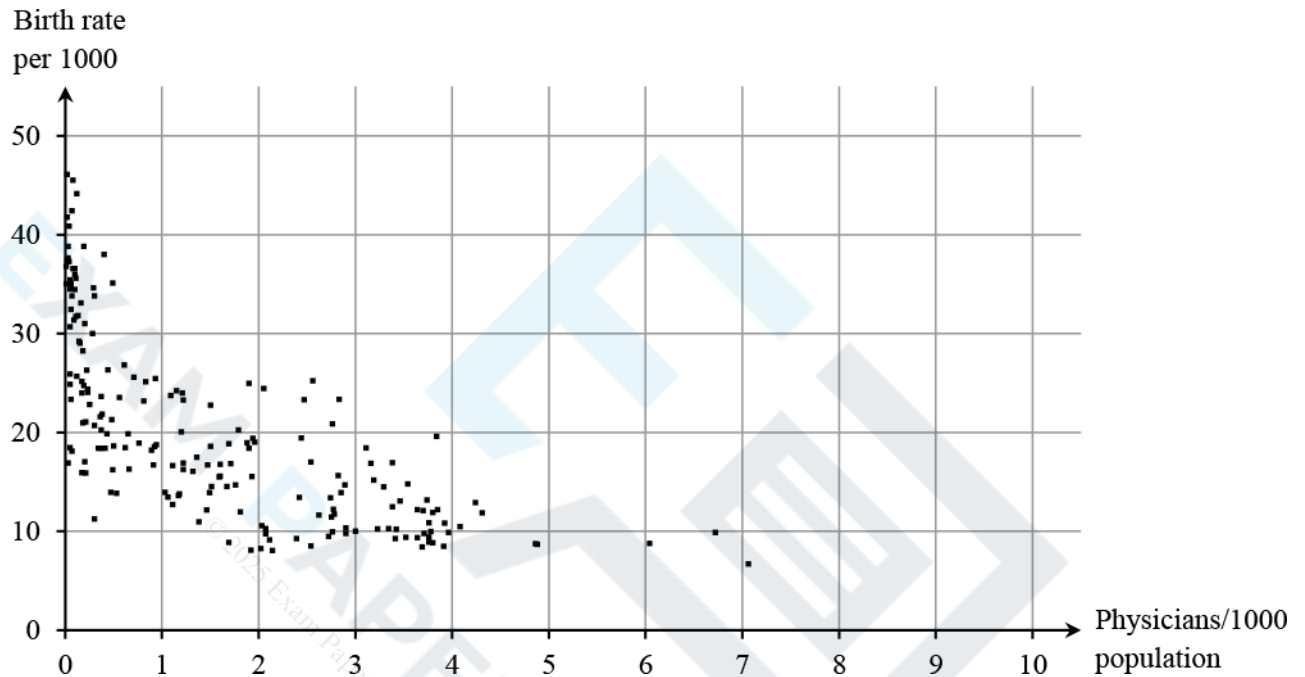
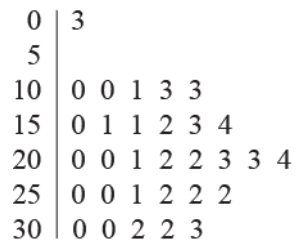


Fig. 9.2

- (c) Describe the correlation in the scatter diagram. [1]
- (d) Discuss briefly whether the scatter diagram shows that high birth rates would be reduced by increasing the number of physicians in a country. [1]

- 17 The maximum daytime temperature was recorded on each day in May 2016 at a weather station in Canada. The data are shown in the stem-and-leaf diagram below.



key: 15 | 1 represents a temperature of 16 °C

- (a) Describe the shape of the distribution.

[1]

- (b) Find the interquartile range.

[2]

- (c) Hence determine whether 3 is an outlier.

[2]

- 18 A recruitment company advertises vacancies on their website. Information on the salaries for 36 of these vacancies is given in Fig. 7. The data have been grouped.

Salary in thousands of pounds	20 –	25 –	30 –	35 –	40 –	45 –	50 – 55
Number of vacancies	3	6	6	12	3	3	3

Fig. 7

- (a) For these salaries, calculate estimates of
- the mean,
 - the sample standard deviation.

Give your answers to the nearest pound.

[4]

- (b) Explain why your values are only estimates.

[1]

- (c) Give a reason why it would not be appropriate to use the mean calculated in part (a) as an estimate of the mean salary for all vacancies in the country.

[1]

- (d) Another vacancy has an annual salary of £52 573. This was not included in the table. Without further calculation, state how the mean salary would be affected if it were to be recalculated including this value. [1]

- 19 The managing director of an international internet communications company wishes to investigate the number of internet users and the number of mobile phone users for different countries in Eastern Europe. He wishes to identify a country with the potential to increase the number of internet users so that he can consider investing in that country.

Fig. 11.1 shows all the countries of Eastern Europe.

Country (in Eastern Europe)	Mobile phone subscribers (millions)	Internet users (millions)
Albania	3.50	1.30
Belarus	10.68	2.64
Bosnia and Herzegovina	3.35	1.42
Bulgaria	10.78	3.40
Croatia	4.97	2.23
Czech Republic	12.97	6.68
Estonia	2.07	0.97
Hungary	11.58	6.18
Kosovo	0.56	missing
Moldova	4.08	1.33
Montenegro	1.13	0.28
Poland	50.84	22.45
Romania	22.70	7.79
Serbia	9.14	4.11
Slovakia	6.10	4.06
Slovenia	2.25	1.30
Ukraine	59.34	7.77

Source: CIA World Factbook

Fig. 11.1

- (a) Are the countries in Fig. 11.1 a sample or a population? Explain your answer.

[1]

- (b) These data have been used to construct the scatter diagram in Fig. 11.2. Use your knowledge of the large data set to comment on the correlation in the scatter diagram.

[2]

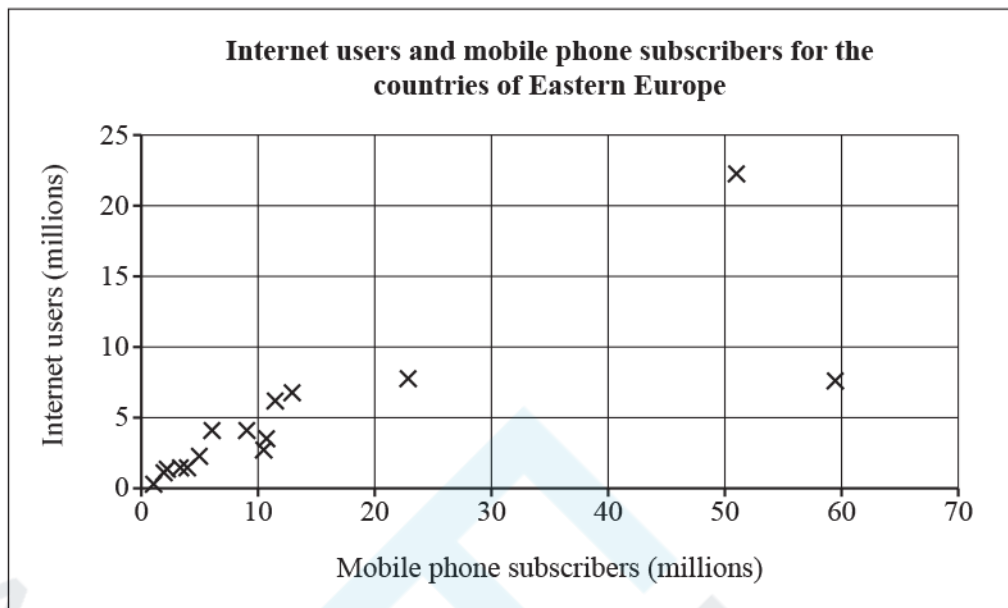


Fig. 11.2

- (c) Use the scatter diagram to identify a country that appears to have high potential to increase the number of internet users. Give a reason for your choice.

[2]

- (d) Having decided on the country he wishes to invest in, the director will select a sample of 20 marketing consultants from that country to contact for information and advice. He has found a website listing 600 marketing consultants. Give clear instructions for the director on how to select a simple random sample of 20 marketing consultants from the 600. [3]

20 The head of sales of a large company presented Fig. 3 to the board of directors as part of his end-of-year report.

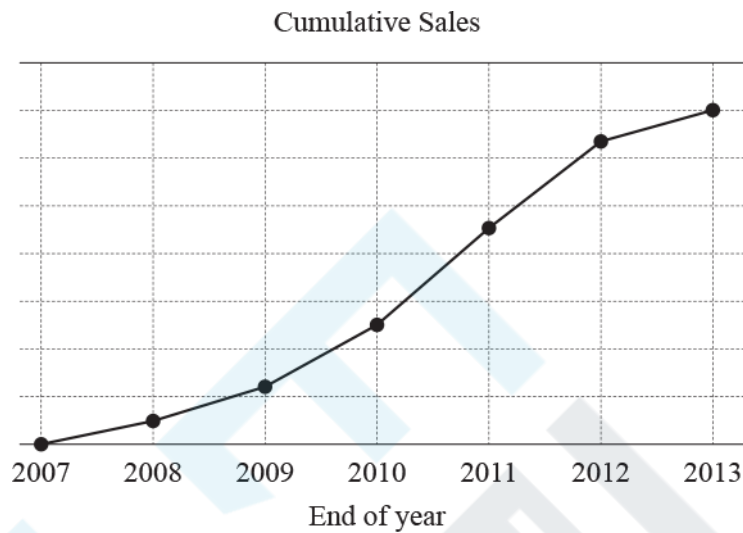


Fig. 3

- (a) What key feature is missing from this graph? [1]

One director comments that the diagram shows increasing year-on-year sales.

- (b) Explain whether the director is correct. [1]

- 21 The screenshot in Fig. 8 shows the probability distribution for the discrete random variable X , where $X \sim B(20, 0.9)$.

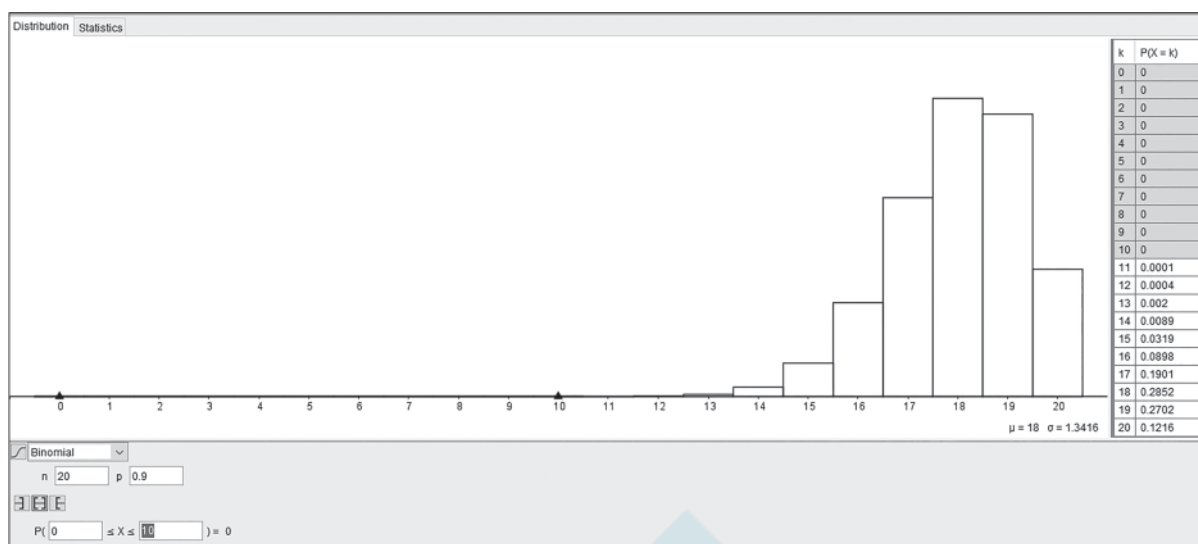


Fig. 8

- (a) Describe the shape of the distribution. [1]
- (b) Explain why the values of $P(X = k)$ for $k = 0$ to 10 inclusive are recorded as 0 in the table in the screenshot. [1]
- (c) State which of the values from 0 to 20 is
- (i) the most likely value of X , [1]
- (ii) the least likely value of X . [1]

- 22 (a) (i) The pre-release data shows that the total population of the 239 countries in the world in 2016 was 7174654290. The populations of a sample of 10 countries are given in Fig. 9.1.

Country	Population
Tuvalu	10 782
Equatorial Guinea	722 254
Somalia	10 428 043
Denmark	5 569 077
Burma	55 746 253
Norway	5 147 792
Botswana	2 155 784
Rwanda	12 337 138
Sint Maarten	39 689
Swaziland	1 419 623

Fig. 9.1

Show that the mean population per country for the whole world is much larger than the mean population per country for this sample.

[3]

- (ii) Rebecca takes a large number of different samples of 10 countries. She finds that the mean population per country is usually smaller for the sample than it is for the whole world. Explain whether this suggests that the sampling was not random.

[2]

- (b) Fig. 9.2 shows data for Norway.

Country	Population	GDP per capita (US\$)	Health expenditure (% of GDP)
Norway	5 147 792	55 400	9.1

Fig. 9.2

Calculate Norway's health expenditure per person in US\$.

[2]

- (c) As part of an investigation into the factors which might be associated with life expectancy, the scatter diagrams in Fig. 9.3 are drawn. The line of best fit and the corresponding value of the square of the correlation coefficient are also shown for each scatter diagram.

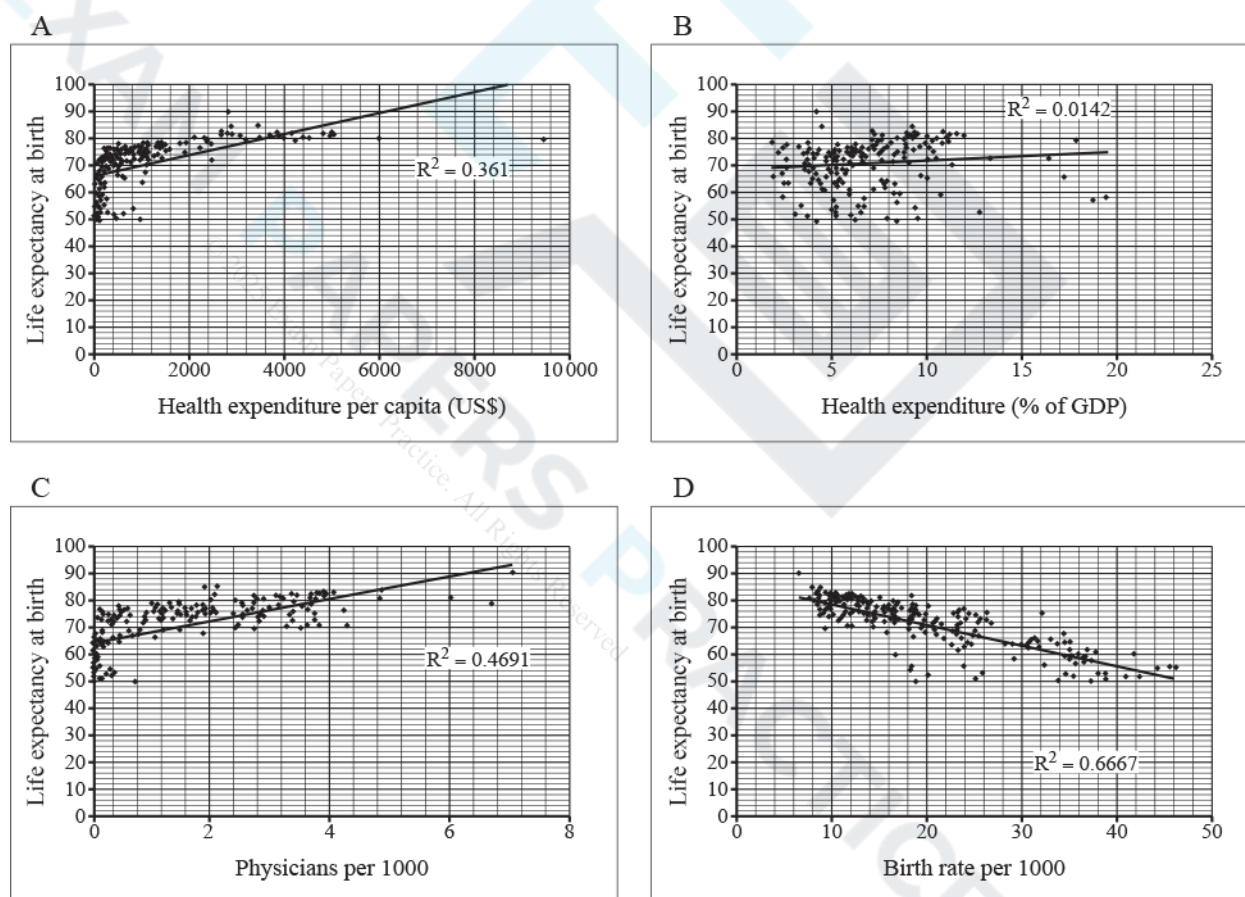


Fig. 9.3

- (i) Which of the four factors appears to have the strongest positive association with life expectancy at

birth? Give a reason for your answer.

[2]

- (ii) Explain why the line of best fit in scatter diagram B is not a good model for the relationship between the two variables.

[1]

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- 23 The numbers of units of electricity, x kWh (kilowatt-hours), used by 50 customers of an energy firm in a period of one month are summarised as follows.

$$\Sigma x = 17\,100$$

$$\Sigma x^2 = 6115\,108$$

- (i) Calculate the mean and standard deviation of x .

[3]

- (ii) The cost, $\pounds y$, of the electricity used by each customer is given by the formula $y = 0.108x + 7.2$. Use your answers to part (i) to deduce the mean and standard deviation of the costs of the electricity used by these customers.

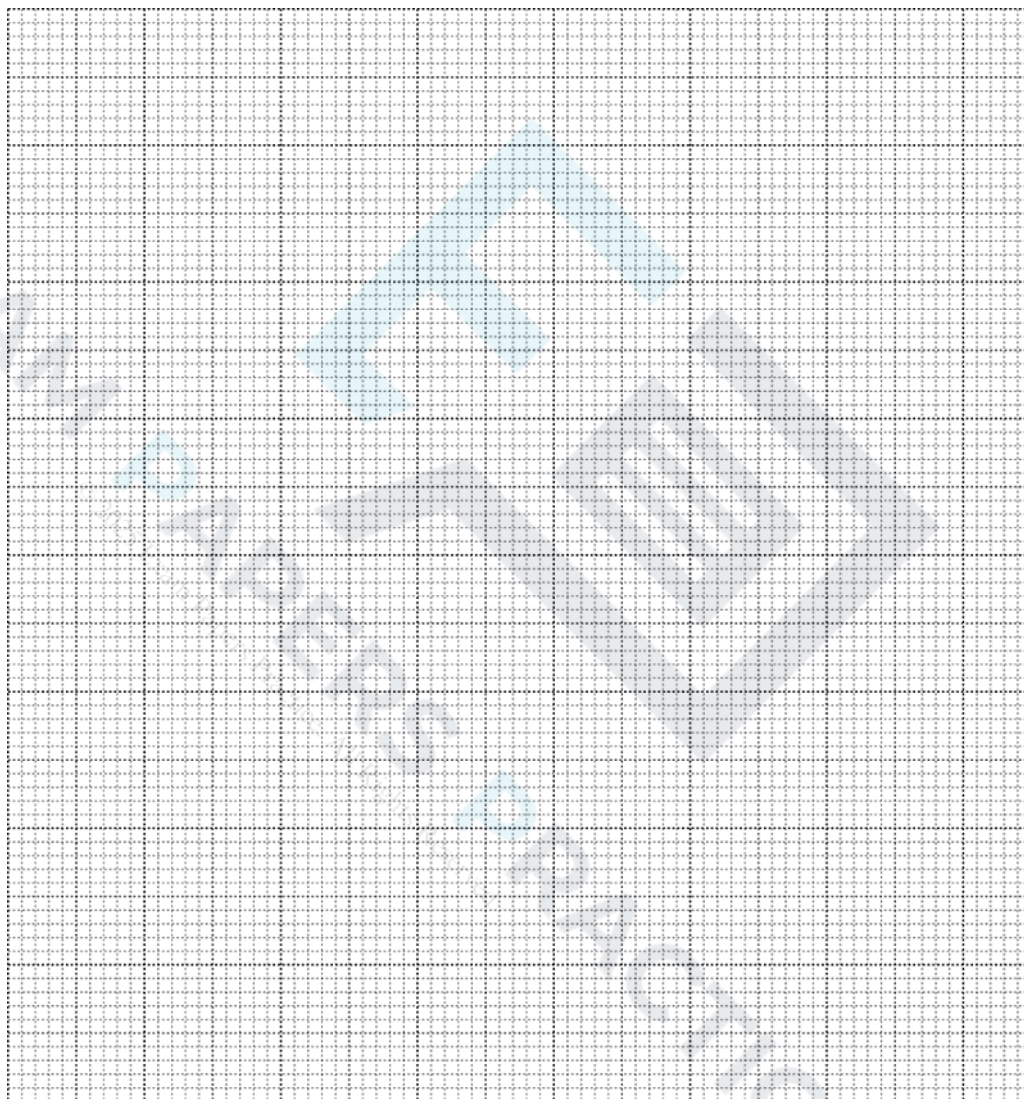
[3]

- 24 The table below shows the maximum daily level of the pollutant nitrogen dioxide in Marylebone Road in London in 2015. The levels are measured in micrograms per cubic metre ($\mu\text{g} / \text{m}^3$). There were 7 days where no figures were available.

Pollutant level ($x \mu\text{g}/\text{m}^3$)	$40 \leq x < 80$	$80 \leq x < 120$	$120 \leq x < 140$	$140 \leq x < 180$	$180 \leq x < 220$	$220 \leq x \leq 300$
Frequency	29	74	52	129	64	10

- (i) Draw a cumulative frequency diagram to illustrate the data.

[5]



- (ii) Levels of nitrogen dioxide below 200 are classified as low. Estimate the proportion of days on which the level was low.

[2]

- (iii) Use your diagram to estimate the median and interquartile range of the data.

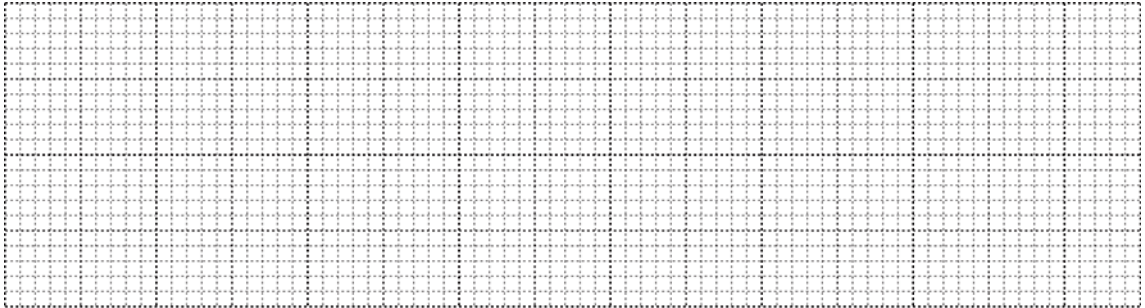
[3]

- (iv) For each end of the distribution, explain whether outliers definitely exist, may possibly exist or definitely do not exist.

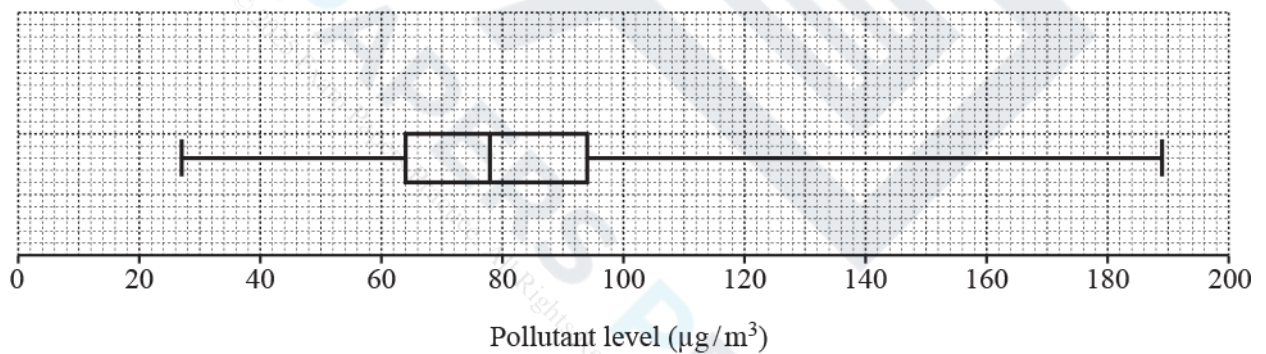
[4]

(v) Draw a box and whisker plot to illustrate the data.

[3]



The box and whisker plot below shows similar data for a roadside location in Tower Hamlets in London.



(vi) Compare the skewness of the data from the two locations.

[2]

25 The probability distribution for the discrete random variable X is shown in Fig. 3.

x	0	1	2	3	4
$P(X = x)$	$0.4a$	$0.5a$	$0.3a$	$0.2a$	$0.1a$

Fig. 3

(a) Calculate the value of the constant a .

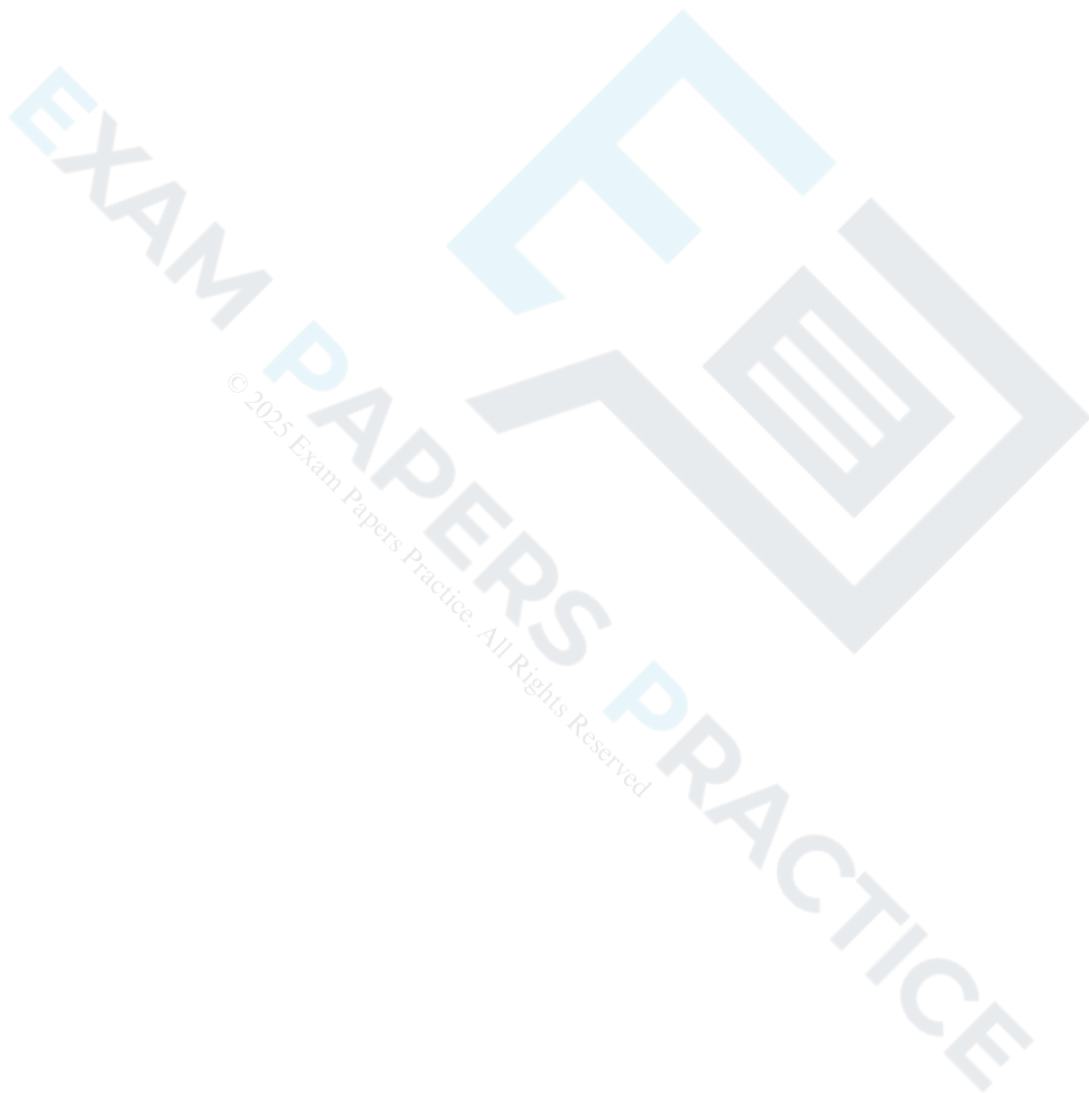
[2]

(b) Calculate $P(X \geq 2)$.

[2]

(c) Describe the shape of the distribution.

[1]



26 Nigel investigated migration in and out of different areas in the United Kingdom. He considered all the data available for long term international inflow and outflow rates per thousand resident population in 2012. Fig. 9.1 shows a scatter diagram for Nigel's sample after the data had been cleaned.

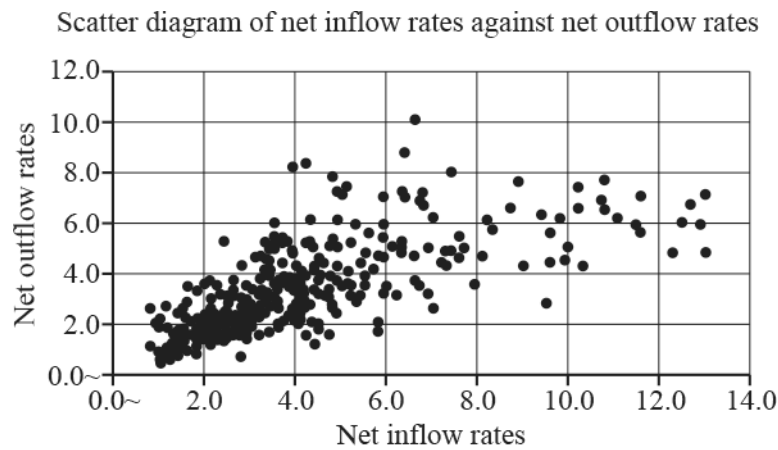


Fig. 9.1

Nigel used software to generate summary statistics. Fig. 9.2 shows summary statistics for inflow rates.

Statistics	
n	391
mean	6.8
s	8.579
$\sum x$	2644.9
$\sum x^2$	46598.17
min	0.8
Q_1	2.5
median	4.0
Q_3	6.8
max	91.8

Fig. 9.2

Nigel decided to clean the data by discarding all outliers, and so the pairs of values with long term inflow rates greater than 13.25 were discarded.

(a) Use information from Fig. 9.2 to show how Nigel calculated the value 13.25. [1]

Nigel then adopted a similar procedure by considering summary statistics for the long term net outflow rates. As a result he obtained a sample of size 334 drawn from a population of 391.

(b) State **one** strength and **two** weaknesses of Nigel's method for selecting his sample.

[3]



Nigel found that the product moment correlation coefficient for these data is 0.6916. He used an online statistics calculator to find the associated p -value. A screenshot of the result is shown in Fig. 9.3.

Calculate One, Two Tailed P-Value Correlation Probability

Enter r Value
0.6916

Enter V Value
334

Calculate Reset

t
17.446889221076393

df
332

P (One-Tailed)
0.000000

P (Two-Tailed)
0.000000

Fig. 9.3

Nigel made the following statement.

“My analysis proves that there is positive correlation between long term net inflow and long term net outflow rates per thousand resident population. Therefore high long term inflow rates cause high long term outflow rates.”

(c) Give **three** reasons why Nigel's statement is flawed.

[3]

27 The pre-release data set consists of a subset (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>) of the data about countries from the CIA World Facebook.

Data for Croatia shows that the average number of mobile phone subscribers per 1000 population is 1111.72.

(a) Explain how this average can be greater than 1000.

[1]

It is assumed that, once the data have been cleaned, the average number of phones per 1000 population in different countries can reasonably be approximated by a Normal distribution with mean 996 and standard deviation 407.

(b) State two ways in which the data may have been cleaned.

[2]

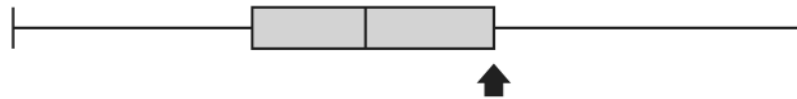


Fig. 11.1

Fig. 11.1 shows a box plot of the cleaned data.

(c) Give the name of the measure which the arrow is pointing to.

[2]

(d) Use the approximate Normal distribution to estimate a value for this measure.

[2]

Fatima thinks that the mean number of mobile phone subscribers per 1000 population for the countries of the world has increased since the data were collected. In order to test this he obtains some up-to-date data for a random sample of countries and uses software to conduct the appropriate hypothesis test at the 5% level of significance. The output from the software is shown in Fig. 11.2.

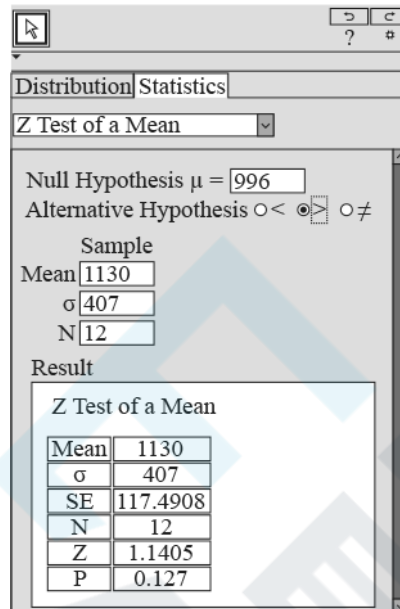


Fig. 11.2

(e) State the conclusion of Fatima's test, explaining your reasoning.

[4]

(f) With reference to the pre-release data set, comment on the sample size Fatima has chosen.

[1]

- 28 A team of biologists are monitoring a subspecies of vole on a remote island. Previous studies have shown that the population mean mass of adult male voles is 26.2 g. It is believed that, due to changes in the environment, the mean mass may have altered. Three of the biologists conducted a new survey and each collected a random sample of adult male voles. The voles were weighed and tagged and then released unharmed.

Fig. 12 shows summary statistics for the 3 samples.

Statistics	Sample 1	Sample 2	Sample 3
Mean	22.6	22.8	23.7
Standard deviation	2.983287	3.258527	4.122196
Variance	8.9	10.618	16.9925
Minimum	19.2	18.6	18.4
Maximum	29.1	30.1	32
Q_1	20.3	20.1	19.8
Median	21.7	22.4	23.2
Q_3	23.95	24.3	26.1
Count	9	11	17

Fig. 12

- (a) For each sample calculate Σx and Σx^2 .

[4]

- (b) Use your answers to part (a) to calculate the mean and variance of the mass of all the voles caught in the survey. [2]

The biologists use the results of the survey to test, at the 5% level, the hypothesis $H_0: \mu = 26.1$ against the hypothesis $H_1: \mu \neq 26.1$.

- (c) Assuming that the distribution of the masses of adult male voles may be modelled by the Normal distribution with the value for the variance found in part (b), calculate the critical region for this test. [3]

(d) State the conclusion reached by the biologists, explaining your answer.

[1]

- 29 Doug has a list of times taken by competitors in a 'fun run'. He has grouped the data and calculated the frequency densities in order to draw a histogram to represent the information. Some of the data are presented in Fig. 2.

Time in minutes	15 –	20 –	25 –	35 –	45 – 60
Number of runners	12	23	59	71	
Frequency density	2.4		5.9	7.1	1.4

Fig. 2

(a) Write down the missing values in Fig. 2 above.

[2]

- (b) Doug labels the horizontal axis on the histogram 'time in minutes' and the vertical axis 'number of minutes per runner'. State which one of these labels is incorrect and write down a correct version. [1]

- 30 Rose and Emma each wear a device that records the number of steps they take in a day. All the results for a 7-day period are given in Fig. 7.

Day	1	2	3	4	5	6	7
Rose	10 014	11 262	10 149	9361	9708	9921	10 369
Emma	9204	9913	8741	10 015	10 261	7391	10 856

Fig. 7

The 7-day mean is the mean number of steps taken in the last 7 days. The 7-day mean for Rose is 10 112.

- (a) Calculate the 7-day mean for Emma. [1]

At the end of day 8 a new 7-day mean is calculated by including the number of steps taken on day 8 and omitting the number of steps taken on day 1. On day 8 Rose takes 10 259 steps.

- (b) Determine the number of steps Emma must take on day 8 so that her 7-day mean at the end of day 8 is

the same as for Rose.

[4]

In fact, over a long period of time, the mean of the number of steps per day that Emma takes is 10 341 and the standard deviation is 948.

- (c) Determine whether the number of steps Emma needs to take on day 8 so that her 7-day mean is the same as that for Rose in part (b) is unusually high.

[3]

- 31 The pre-release material contains data concerning the death rate per thousand people and the birth rate per thousand people in all the countries of the world. The diagram in Fig. 11.1 was generated using a spreadsheet and summarises the birth rates for all the countries in Africa.

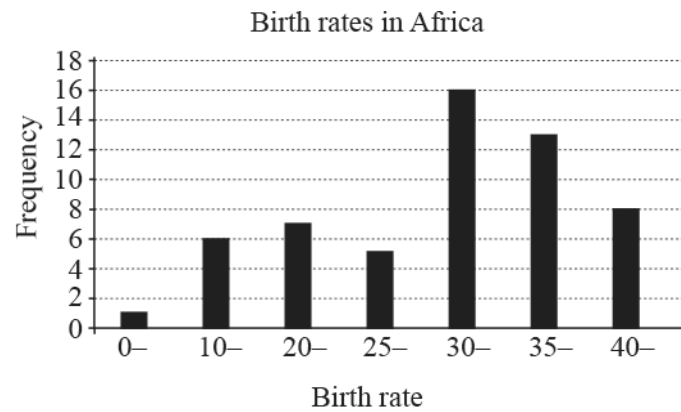


Fig. 11.1

- (a) Identify **two** respects in which the presentation of the data is incorrect.

[2]

Fig. 11.2 shows a scatter diagram of death rate, y , against birth rate, x , for a sample of 55 countries, all of which are in Africa. A line of best fit has also been drawn.

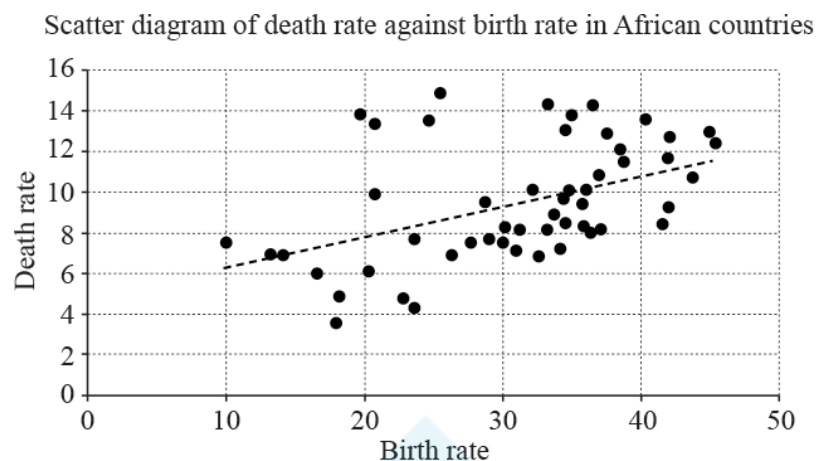


Fig. 11.2

The equation of the line of best fit is $y = 0.15x + 4.72$.

(b) (i) What does the diagram suggest about the relationship between death rate and birth rate? [1]

(ii) The birth rate in Togo is recorded as 34.13 per thousand, but the data on death rate has been lost. Use the equation of the line of best fit to estimate the death rate in Togo. [1]

- (iii) Explain why it would not be sensible to use the equation of the line of best fit to estimate the death rate in a country where the birth rate is 5.5 per thousand. [1]

- (iv) Explain why it would not be sensible to use the equation of the line of best fit to estimate the death rate in a Caribbean country where the birth rate is known. [1]

(v) Explain why it is unlikely that the sample is random.

[1]

Including Togo there were 56 items available for selection.

(c) Describe how a sample of size 14 from this data could be generated for further analysis using systematic sampling.

[2]

32 A survey of the number of cars per household in a certain village generated the data in Fig. 4.

Number of cars	0	1	2	3	4
Number of households	8	22	31	27	7

Fig. 4

(a) Calculate the mean number of cars per household.

[1]

(b) Calculate the standard deviation of the number of cars per household.

[1]

- 33 At the end of each school term at North End College all the science classes in year 10 are given a test. The marks out of 100 achieved by members of set 1 are shown in Fig. 9.

3		5
4		0 9
5		2 3 6
6		0 1 3 5 6
7		0 1 2 5 6 8 9 9
8		3 4 6 6 8 8 9
9		5 5 5 6 7

Key 5 | 2 represents a mark of 52

Fig. 9

- (a) Describe the shape of the distribution. [1]

- (b) The teacher for set 1 claimed that a typical student in his class achieved a mark of 95. How did he justify this statement? [1]

- (c) Another teacher said that the average mark in set 1 is 76. How did she justify this statement? [1]

Benson's mark in the test is 35. If the mark achieved by any student is an outlier in the lower tail of the distribution, the student is moved down to set 2.

- (d) Determine whether Benson is moved down to set 2. [2]

- 34 The pre-release material includes data on unemployment rates in different countries. A sample from this material

has been taken. All the countries in the sample are in Europe. The data have been grouped and are shown in Fig 14.1.

Unemployment rate	0–	5–	10–	15–	20–	35–50
Frequency	15	21	5	5	2	2

Fig. 14.1

A cumulative frequency curve has been generated for the sample data using a spreadsheet. This is shown in Fig. 14.2.

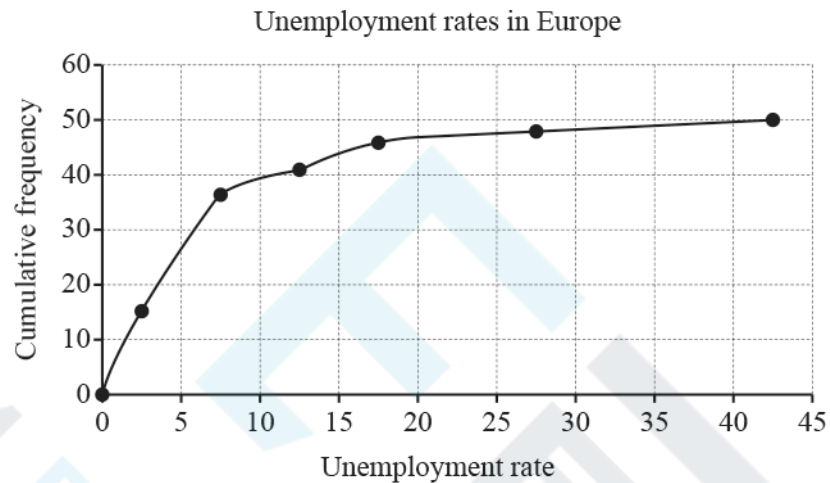


Fig. 14.2

Hodge used Fig. 14.2 to estimate the median unemployment rate in Europe. He obtained the answer 5.0. The correct value for this sample is 6.9.

(a) (i) There is a systematic error in the diagram.

- Identify this error.
- State how this error affects Hodge's estimate.

[2]

(ii) There is another factor which has affected Hodge's estimate.

- Identify this factor.
- State how this factor affects Hodge's estimate.

[2]

(b) Use your knowledge of the pre-release material to give another reason why any estimation of the median unemployment rate in Europe may be unreliable.

[1]

(c) Use your knowledge of the pre-release material to explain why it is very unlikely that the sample has been randomly selected from the pre-release material.

[1]

The scatter diagram shown in Fig. 14.3 shows the unemployment rate and life expectancy at birth for the 47 countries in the sample for which this information is available.

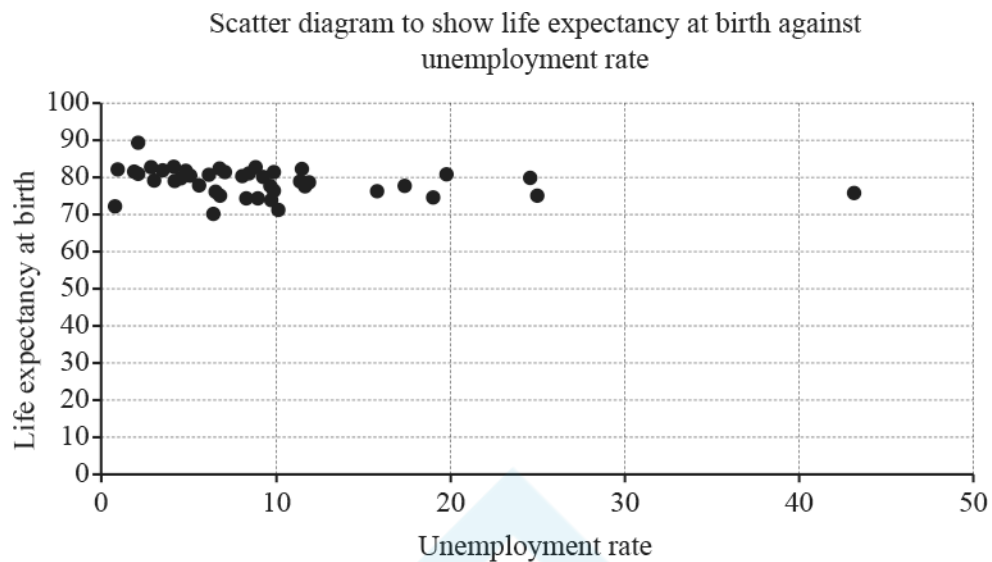


Fig. 14.3

The product moment correlation coefficient for the 47 items in the sample is -0.2607 .

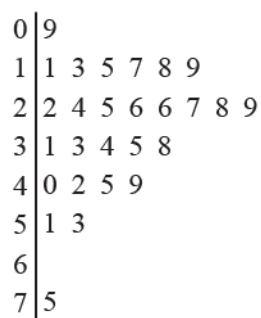
The p -value associated with $r = -0.2607$ and $n = 47$ is 0.0383.

- (d) Does this information suggest that there is an association between unemployment rate and life expectancy at birth in countries in Europe? [2]

Hodge uses the spreadsheet tools to obtain the equation of a line of best fit for this data.

- (e) The unemployment rate in Kosovo is 35.3, but there is no data available on life expectancy. Is it reasonable to use Hodge's line of best fit to estimate life expectancy at birth in Kosovo? [1]

- 35 The stem-and-leaf diagram in Fig. 7 shows the numbers of customers at a village post office on the days it was open in March 2017.



Key 4|0 represents 40 customers

Fig. 7

- (a) Describe the shape of the distribution.

[1]

- (b) Draw a box plot to represent the data.

[4]



- 36 At the start of the January term year 11 students at Amplesides College sat mock examinations in GCSE

mathematics papers 4 and 5. A teacher collected the results and presented them in a scatter diagram, which is shown in Fig. 10. A line of best fit has been added.

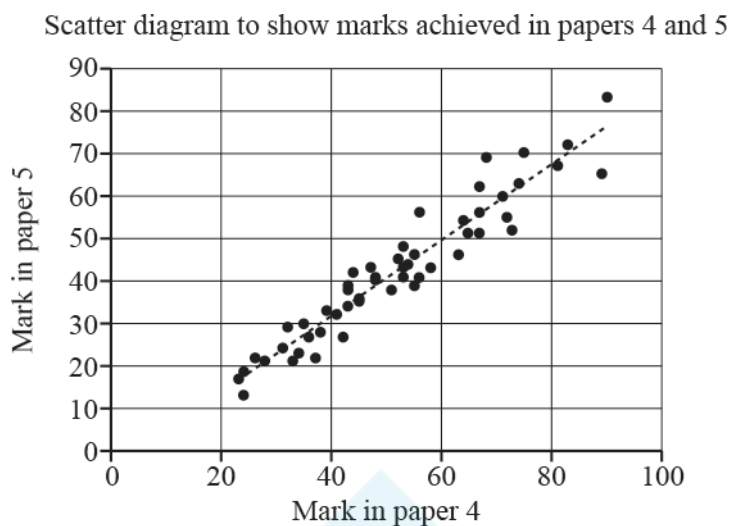


Fig. 10

The correlation coefficient for the data is 0.9566.

- (a) Give **two** reasons why it is reasonable to model the relationship between the mark achieved in paper 4 and the mark achieved in paper 5 by a straight line. [2]

The equation of the line of best fit is $y = 0.89x - 3.76$, where y is the mark achieved in paper 5 and x is the mark achieved in paper 4.

- (b) Tina achieved a mark of 83 in paper 4, but was absent for paper 5. Calculate an estimate of the mark she would have achieved in paper 5. [1]

- (c) Dave was absent for paper 4. He achieved a mark of 8 in paper 5. Calculate an estimate of the mark Dave would have achieved in paper 4. [2]

- (d) Explain why the estimate of Tina's mark for paper 5 is more reliable than the estimate of Dave's mark for paper 4. [1]

- 37 The spreadsheet output in Fig. 12.1 gives some information about all the countries that won more than 10 gold medals in the London 2012 Olympic Games.

A	B	C	D	E	F	G	H
1	Country	population	GDP per capita (US\$)	Total GDP (US\$)	Gold medals	Silver medals	Bronze medals
2	United States	318 892 103	52 800	1.68×10^{13}	46	29	29
3	China	1 355 692 576	9800	1.33×10^{13}	38	27	22
4	United Kingdom	63 742 977	37 300	2.38×10^{12}	29	17	19
5	Russia	142 470 272	18 100	2.58×10^{12}	24	25	33
6	Korea, South	49 039 986	33 200	1.63×10^{12}	13	8	7
7	France	66 259 012	35 700	2.37×10^{12}	11	11	12
8	Germany	80 996 685	39 500	3.20×10^{12}	11	19	14

Fig. 12.1

- (a) Give a spreadsheet formula for calculating the value in cell D2 using other cell values in Fig. 12.1. [1]

The statistics in Fig. 12.2 are for all the countries in the pre-release data.

	Population	GDP per capita (US\$)	Total GDP (US\$)
Lower Quartile	4.587×10^5	4525	2.09×10^9
Median	5.623×10^6	13750	7.73×10^{10}
Upper Quartile	2.116×10^7	31750	6.72×10^{10}

Fig. 12.2

- (b) Explain whether or not the statements below are consistent with the information given in Fig. 12.1 and Fig 12.2.

Statement A

Countries with larger populations are more likely to win Olympic gold medals.

Statement B

Countries with larger total GDP are more likely to win Olympic gold medals.

[3]

There were approximately 10 500 Olympic competitors in 2012. The population of the world was approximately 7 000 000 000 in 2012.

A geography student assumed that Olympic competitors are randomly and uniformly scattered across the population of the world. The student calculated that the population of a country with two Olympic competitors would be approximately 1 300 000.

- (c) Use your knowledge of the large data set to explain whether the geography student's assumption is realistic.

[1]

- 38 Rose packs eggs in boxes of 6, which she then sells at her farm. During the process some eggs are cracked, and Rose randomly selects a sample of boxes and records the number of cracked eggs in each box. The results are summarised in Fig. 16.

Number of cracked eggs	0	1	2	3	4	5	6
Number of boxes	163	103	28	5	0	0	1

Fig.16

- (a) Calculate the mean number of cracked eggs per box.

[1]

Rose believes that the number of cracked eggs per box may be modelled by a binomial distribution.

- (b) State a modelling assumption that is necessary for a binomial distribution to be used to model the number of cracked eggs per box.

[1]

Rose defines p as the probability that a particular egg is cracked.

(c) Use your answer to part (a) to find the value of p .

[2]

(d) Calculate the expected frequencies of cracked eggs per box according to Rose's model, giving your answers correct to 1 decimal place.

[3]

(e) Comment on whether Rose's model is a good fit for the data.

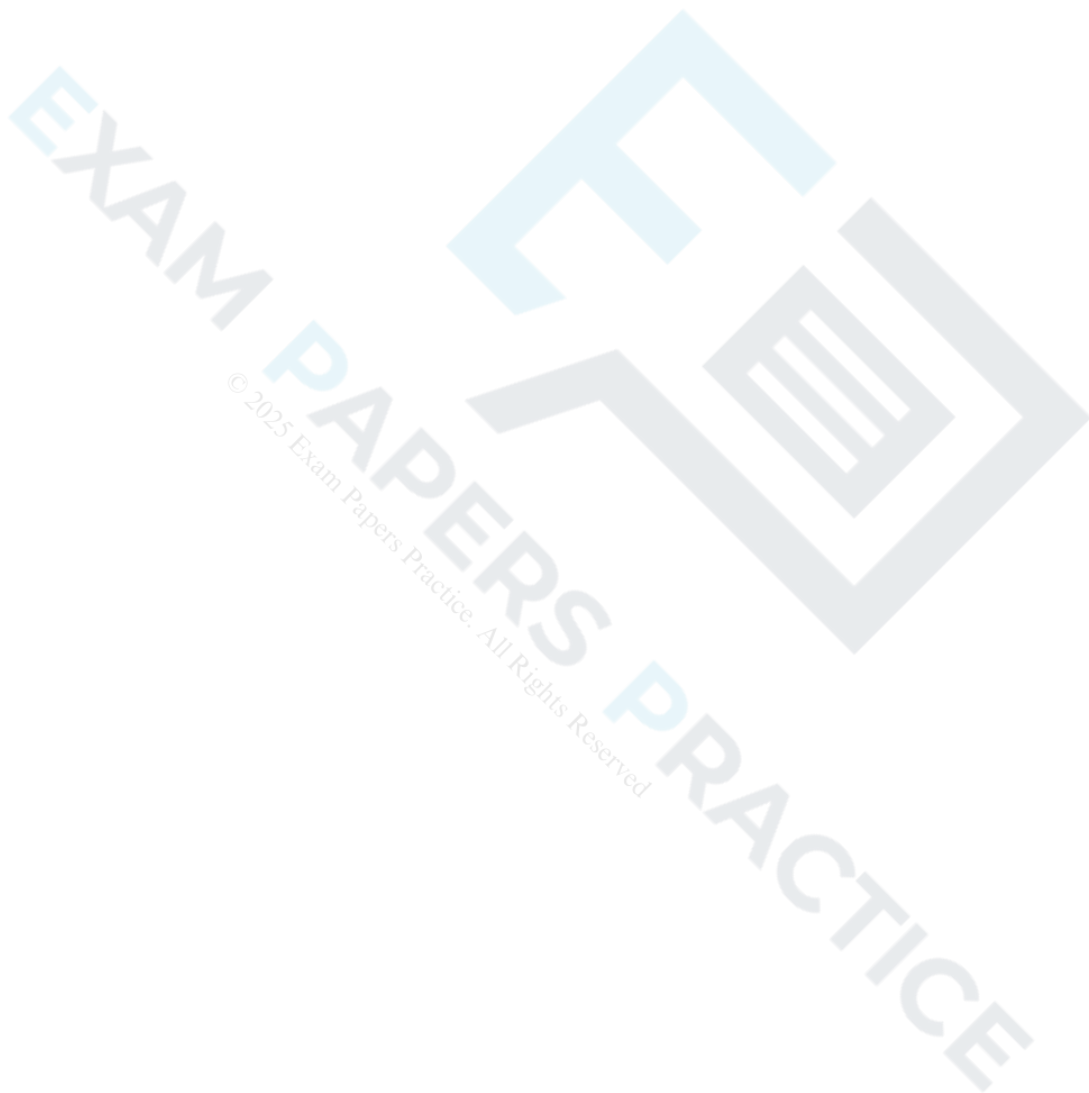
[1]

Instead of selling eggs at the farm, Rose decides to sell them to a wholesaler. The eggs are now selected randomly and packed in open trays of 24. Rose believes that this will result in a change in the probability of an egg being cracked. She selects a tray at random and finds that 5 eggs are cracked.

(f) In this question you must show detailed reasoning.

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that the proportion of cracked eggs has changed.

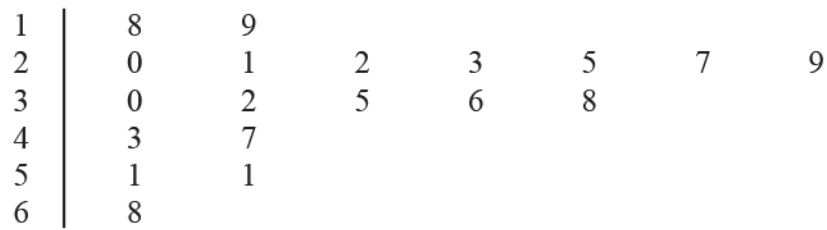
[7]



39 Qasim has just opened a café in Burnton. He decides to conduct some market research on his customers.

EXAM PAPERS PRACTICE

One Monday morning at 11 am he asked every customer in the café to fill out a questionnaire which included asking for the customer's age. The results he collected are shown in the stem-and-leaf diagram in Fig. 7.1.



Key 2 | 0 represents an age of 20.

Fig. 7.1

(a) What name is given to the sampling method used by Qasim? [1]

(b) Explain why the sample does not represent a simple random sample of all the customers who use Qasim's café. [1]

(c) Describe the shape of the distribution of the ages in the sample. [1]

- (d) For the data in Fig. 7.1 find
- the median,
 - the interquartile range.

[3]

Kai works at Qasim's café. He believes that Qasim's sample data may not be representative of all Qasim's customers. One week he asks every customer to fill in the questionnaire. Summary statistics for the customers who filled in the questionnaire are shown in Fig. 7.2.

Number of respondents	237
Age of youngest person	7
Lower quartile	27
Median	31
Upper quartile	39
Age of oldest person	81

Fig. 7.2

- (e) Comment on whether the statistics in Fig. 7.2 provide any evidence to support Kai's belief.

[2]

- 40 For a period of 20 years until the end of 2017 the parish council of the village of Springmoor have met every fortnight to discuss local affairs. Throughout this period Mr. Loparts acted as chairperson.

The parish council secretary has noted that the mean length of time of meetings throughout this period was 142 minutes, and the standard deviation of the length of time of meetings was 17.4 minutes.

At the beginning of 2018 Miss Patel took over as chairperson. At the end of 2018 the secretary stated that he believed that the mean length of time of meetings is lower since Miss Patel became chairperson.

The secretary generated a random sample of lengths of times of meetings in 2018. The times, in minutes, are as follows.

123, 137, 119, 135, 136, 143, 126, 126, 125.

Assume that the length of time of meetings may be modelled by the Normal distribution and that the variance of the length of time of meetings is unaltered.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to support the secretary's belief that the mean length of meeting has decreased.

[7]

- 41 The large data set (LDS1) provides information on average life expectancy at birth for countries of the world. Fig. 13.1 shows the entry for South Sudan, in Africa.

Country	life expectancy at birth
South Sudan	#N /A

Fig. 13.1

No data concerning average life expectancy at birth is available for South Sudan.

- (a) Explain why the spreadsheet entry is #N /A instead of simply being left blank.

[1]

Summary statistics for the values given for average life expectancy at birth for all the countries in Africa apart from South Sudan were generated using software. These are shown in Fig. 13.2.

<i>Mean</i>	61.21418
<i>Standard Deviation (s)</i>	7.83837
<i>Lowest Score</i>	49.81
<i>Highest Score</i>	79.36
<i>Distribution Range</i>	29.55
<i>Total Number of Scores</i>	55
<i>Number of Distinct Scores</i>	54
<i>Lowest Class Value</i>	45
<i>Highest Class Value</i>	79.99
<i>Number of Classes</i>	7
<i>Class Range</i>	5

Fig. 13.2

- (b) Explain why the mean of 61.21418 may not represent a good estimate of the average life expectancy at birth of all people in Africa.

[1]

Fig. 13.3 shows a frequency diagram of average life expectancies of countries in Africa, excluding South Sudan.

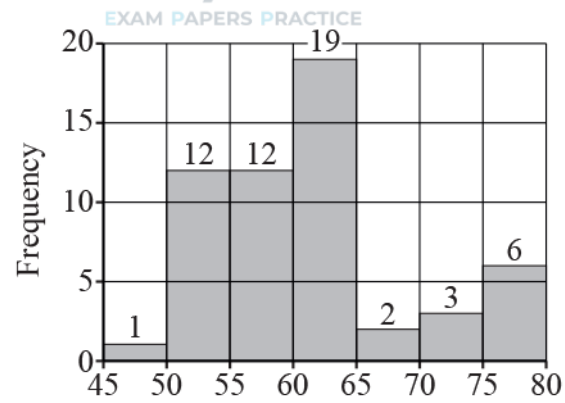


Fig. 13.3

(c) Explain why it would be incorrect to call the diagram in Fig. 13.3 a histogram.

[1]

(d) Draw a histogram to represent the data, using class boundaries at 45, 55, 60, 65 and 80.

[3]

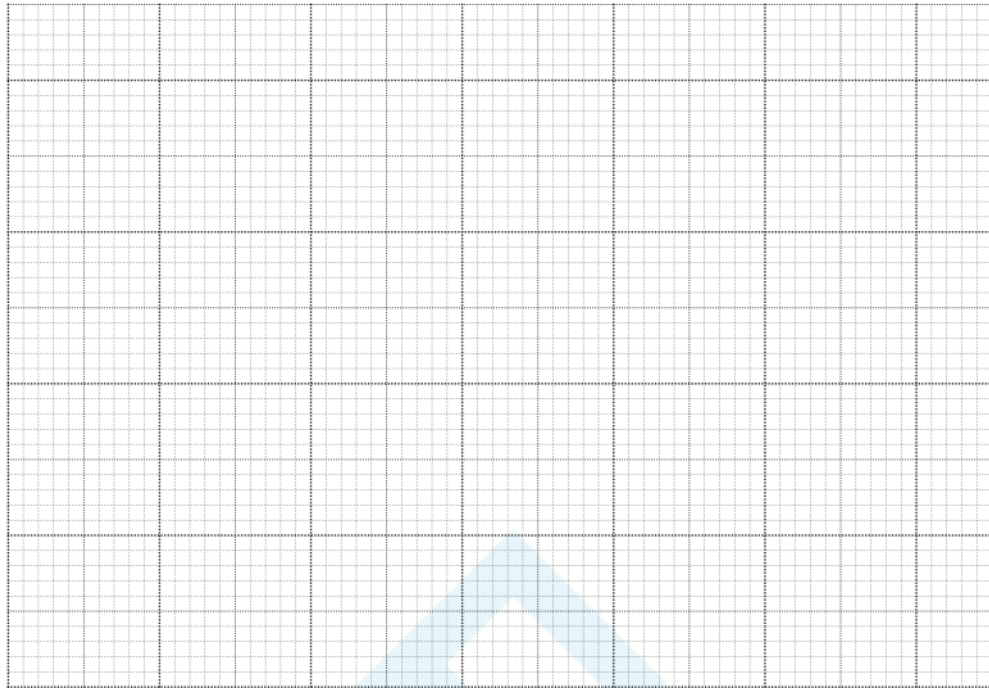


Fig. 13.4 shows a scatter diagram of median age in years against average life expectancy for countries in Africa excluding South Sudan. This was generated using a spreadsheet.

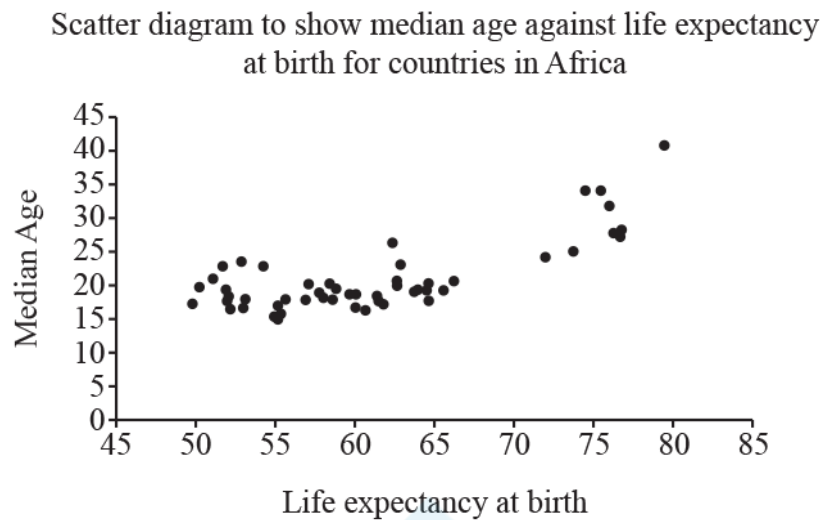


Fig. 13.4

The median age for the population of South Sudan is given as 17.

- (e) With reference to Fig. 13.4, comment on whether it might be possible to obtain a reliable estimate of the average life expectancy at birth for South Sudan.

[2]

42(a) Fig. 3 shows the time Lorraine spent in hours, t , answering e-mails during the working day. The data were collected over a number of months.

Time in hours, t	$0 \leq t < 1$	$1 \leq t < 2$	$2 \leq t < 3$	$3 \leq t < 4$	$4 \leq t < 6$	$6 \leq t < 8$
Number of days	28	36	42	31	24	12

Fig. 3

Calculate an estimate of the mean time per day that Lorraine spent answering e-mails over this period.

[1]

(b) Explain why your answer to the above part is an estimate.

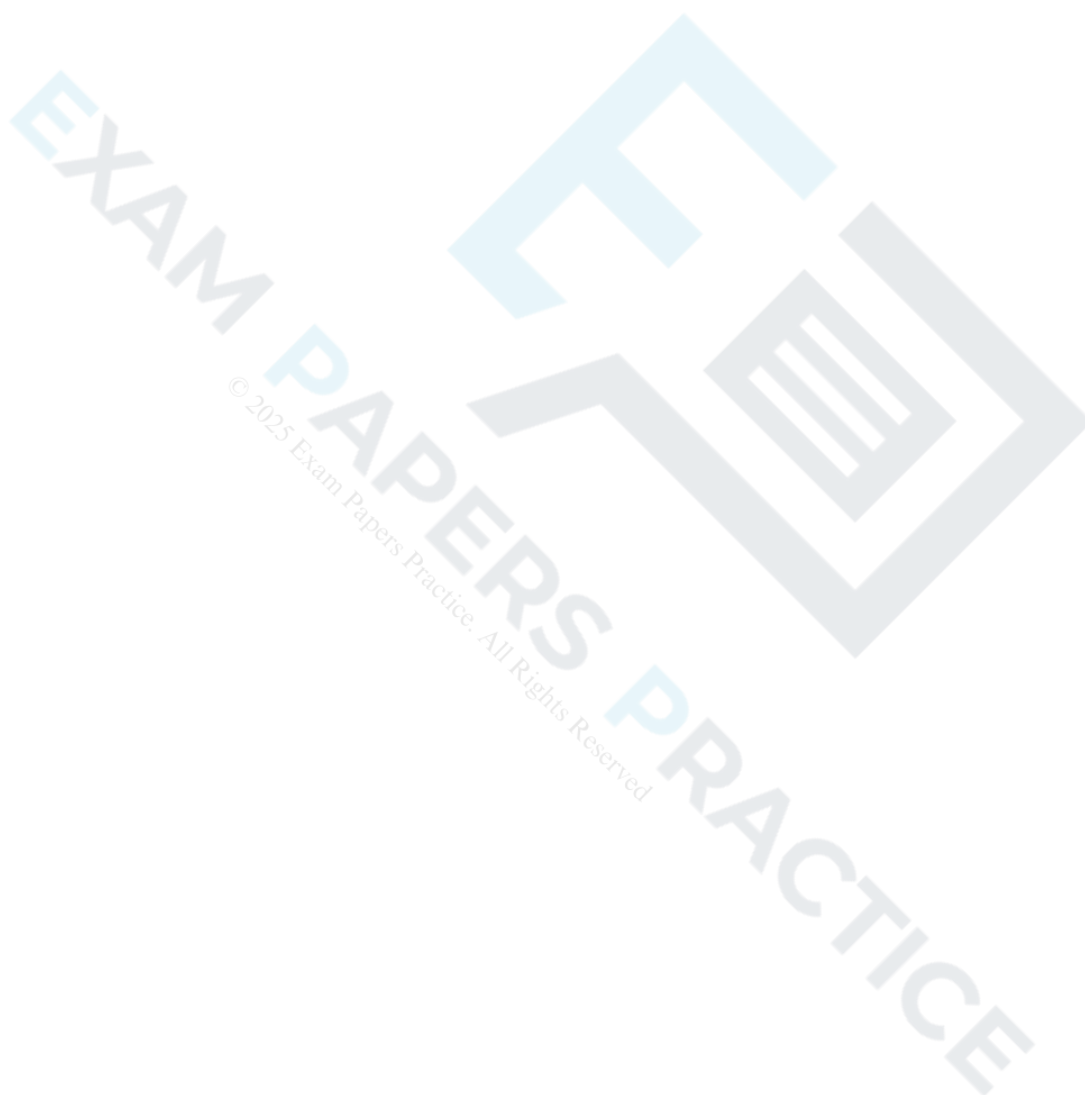
[1]

EXAM PAPERS PRACTICE

(c) When Lorraine accepted her job, she was told that the mean time per day spent answering e-mails would not be more than 3 hours.

Determine whether, according to the data in Fig. 3, it is possible that the mean time per day Lorraine spends answering e-mails is in fact more than 3 hours.

[1]



43(a) Fig. 5 shows the number of times that students at a sixth form college visited a recreational mathematics website during the first week of the summer term.

Number of visits to website	0	1	2	3	4	6
Number of students	24	38	17	12	4	2

Fig. 5

State the value of the mid-range of the data.

[1]

(b) Describe the shape of the distribution.

[1]

(c) State the value of the mode.

[1]

44(a) The pre-release material includes data concerning crude death rates in different countries of the world. Fig. 14.1 shows some information concerning crude death rates in countries in Europe and in Africa.

	Europe	Africa
n	48	56
minimum	6.28	3.58
lower quartile	8.50	7.31
median	9.53	8.71
upper quartile	11.41	11.93
maximum	14.46	14.89

Fig. 14.1

Use your knowledge of the large data set to suggest a reason why the statistics in Fig. 14.1 refer to only 48 of the 51 European countries.

[1]

(b) Use the information in Fig. 14.1 to show that there are no outliers in either data set.

[3]

- (c) The crude death rate in Libya is recorded as 3.58 and the population of Libya is recorded as 6 411 776.

EXAM PAPERS PRACTICE

Calculate an estimate of the number of deaths in Libya in a year.

[1]

- (d) The median age in Germany is 46.5 and the crude death rate is 11.42. The median age in Cyprus is 36.1 and the crude death rate is 6.62.

Explain why a country like Germany, with a higher median age than Cyprus, might also be expected to have a higher crude death rate than Cyprus.

[1]

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- (e) Fig. 14.2 shows a scatter diagram of median age against crude death rate for countries in Africa and Fig. 14.3 shows a scatter diagram of median age against crude death rate for countries in Europe.

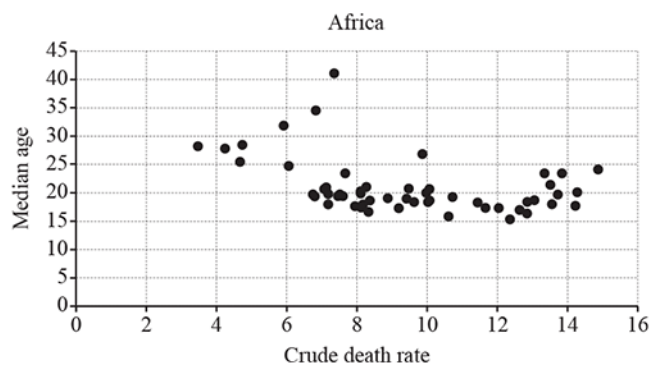


Fig. 14.2

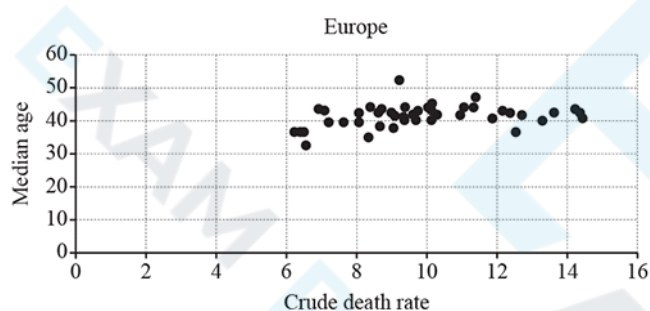


Fig. 14.3

The rank correlation coefficient for the data shown in Fig. 14.2 is -0.281206.

The rank correlation coefficient for the data shown in Fig. 14.3 is 0.335 215.

Compare and contrast what may be inferred about the relationship between median age and crude death rate in countries in Africa and in countries in Europe. [2]

45(a) The large data set gives information about life expectancy at birth for males and females in different London boroughs. Fig. 6.1 shows summary statistics for female life expectancy at birth for the years 2012–2014. Fig. 6.2 shows summary statistics for male life expectancy at birth for the years 2012–2014.

Female Life Expectancy at Birth

n	32
Mean	84.2313
s	1.1563
Σx	2695.4
Σx^2	227078.36
Min	82.1
Q1	83.45
Median	84
Q3	84.9
Max	86.7

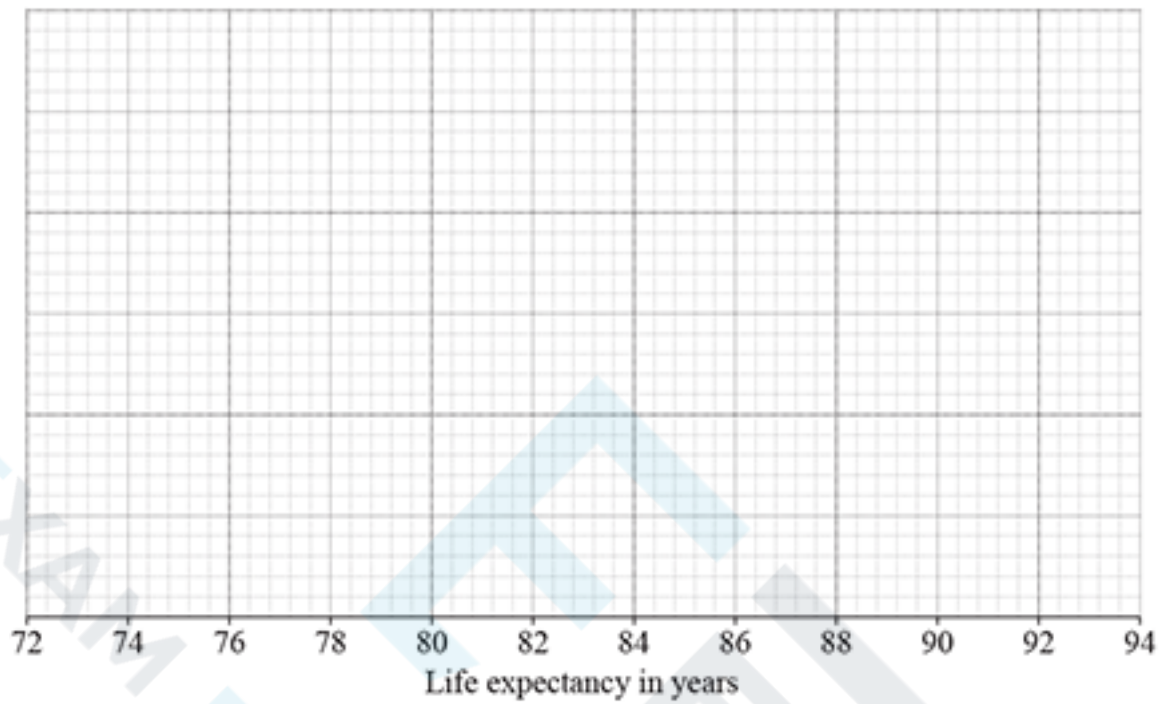
Fig. 6.1

Male Life Expectancy at Birth

n	32
Mean	80.2844
s	1.4294
Σx	2569.1
Σx^2	206321.93
Min	77.6
Q1	79
Median	80.25
Q3	81.15
Max	83.3

Fig. 6.2

Use the information in Fig. 6.1 and Fig. 6.2 to draw two box plots. Draw one box plot for female life expectancy at birth in London boroughs and one box plot for male life expectancy at birth in London



- (b) Compare and contrast the distribution of male life expectancy at birth with the distribution of female life expectancy at birth in London boroughs in 2012–2014.

[2]

- (c) Lorraine, who lives in Lancashire, says she wishes her daughter (who was born in 2013) had been born in the London borough of Barnet, because her daughter would have had a higher life expectancy.

Give **two** reasons why there is no evidence in the large data set to support Lorraine's comment.

[2]

- (d) Use the mean and standard deviation for the summary statistics given in Fig. 6.1 and Fig. 6.2 to show that there is at least one outlier in each set.

[2]

- (e) The scatter diagram in Fig. 6.3 shows male life expectancy at birth plotted against female life expectancy at birth for London boroughs in 2012–14. The outliers have been removed.

Male life expectancy at birth against female life expectancy at birth



Fig. 6.3

Describe the association between male life expectancy at birth and female life expectancy at birth in London boroughs in 2012–14.

[2]

46(a) A student measures the upper arm lengths of a sample of 97 women. The results are summarised in the frequency table in Fig. 2.1.

Arm length in cm	30–	31–	32–	33–	34–	35–	36–	37–	38–	39–	40–41
Frequency	1	4	5	9	13	19	17	17	4	3	5

Fig. 2.1

Upper Arm Length of Women

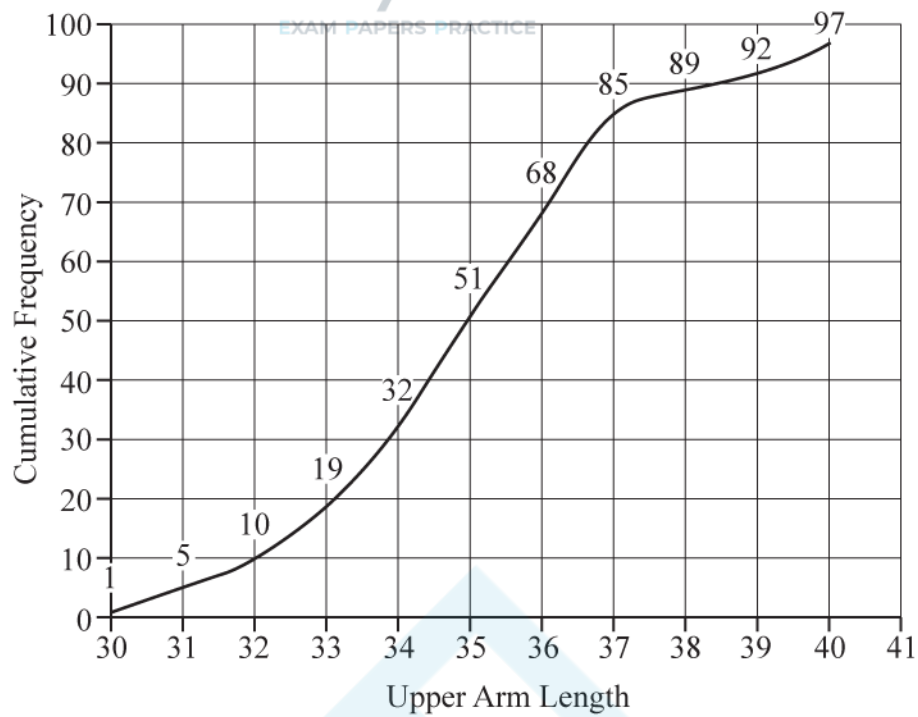


Diagram A

Upper Arm Length of Women

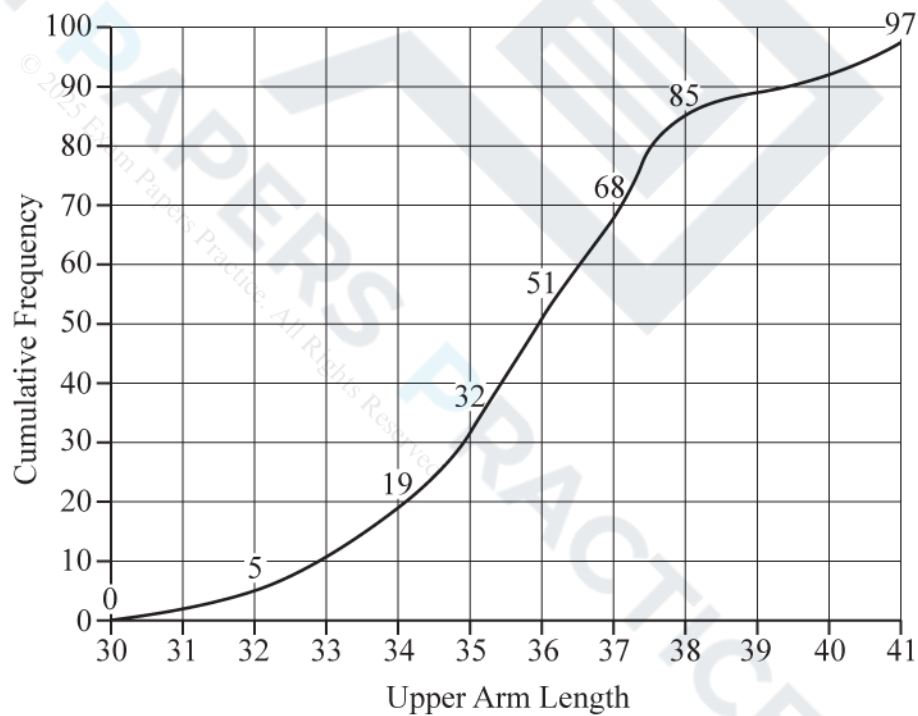


Diagram B

Fig. 2.2

The student constructs two cumulative frequency diagrams to represent the data using different class intervals.

These are shown in Fig. 2.2 above.

One of these diagrams is correct and the other is incorrect.

State which diagram is incorrect, justifying your answer.

[2]

(b) Use the correct diagram in Fig. 2.2 to find an estimate of the median.

[1]

47(a) A researcher is conducting an investigation into the number of portions of fruit adults consume each day. The researcher decides to ask 50 men and 50 women to complete a simple questionnaire.

State the type of sampling procedure the researcher is using.

[1]

(b) Write down one disadvantage of this sampling procedure.

[1]

(c) The researcher represents the data in Fig. 3.1.

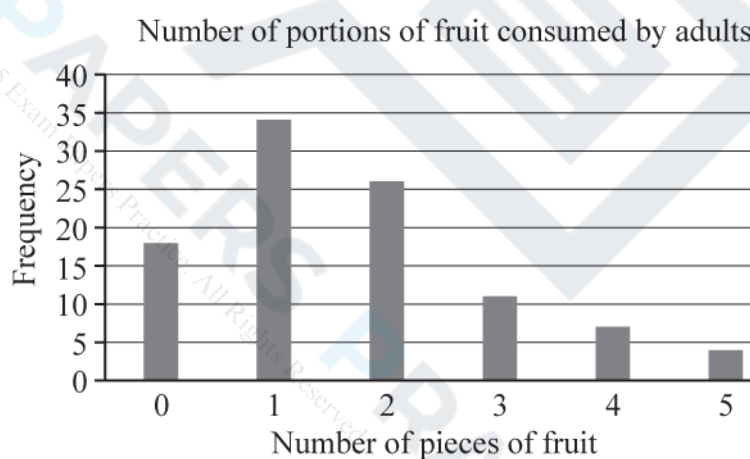


Fig. 3.1

Describe the shape of the distribution.

[1]

(d) The data are summarised in the frequency table in Fig. 3.2.

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Number of portions of fruit	0	1	2	3	4	5
Number of adults	18	34	26	11	7	4

Fig. 3.2

For the data in Fig. 3.2, use your calculator to find

- the mean,
- the standard deviation.

Give your answers correct to 2 decimal places.

[2]

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- (e) A second researcher chooses a proportional stratified sample of 100 children from years 5 and 6 in a certain primary school. There are 220 children to choose from. In year 5 there are 125 children, of whom 81 are boys.

How many year 5 girls should be included in the sample?

[1]

- (f) The second researcher found that the mean number of portions of fruit consumed per day by the children in this sample was 1.61 and the standard deviation was 0.53.

Comment on the amount of fruit consumed per day by the children compared to the amount of fruit consumed per day by the adults.

[2]

48(a) Fig. 10.1 shows a sample collected from the large data set.

BMI is defined as $\frac{\text{mass of person in kilograms}}{\text{square of person's height in metres}^2}$

Sex	Age in years	Mass in kg	Height in cm	BMI
Male	38	77.6	164.8	28.57
Male	17	63.5	170.3	21.89
Male	18	68.0	172.3	22.91
Male	18	57.2	172.2	19.29
Male	19	77.6	191.2	21.23
Male	24	72.7	177.0	23.21
Male	25	92.5	177.9	29.23
Male	26	70.4	159.4	27.71
Male	31	77.5	174.0	25.60
Male	34	132.4	182.2	39.88
Male	38	115.0	186.4	33.10
Male	40	112.1	171.7	38.02

Fig. 10.1

Calculate the mass in kg of a person with a BMI of 23.56 and a height of 181.6 cm, giving your answer correct to 1 decimal place.

[2]

- (b) Fig. 10.2 shows a scatter diagram of BMI against age for the data in the table. A line of best fit has also been drawn.

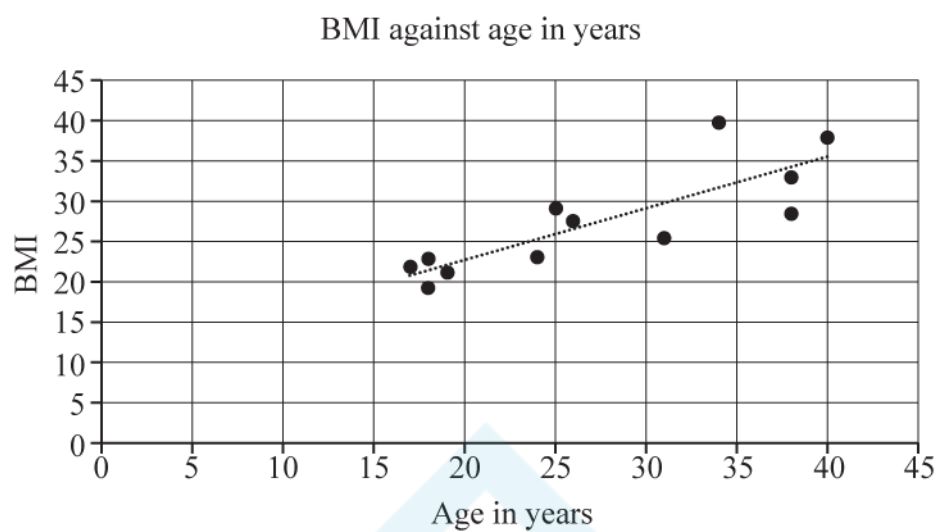


Fig. 10.2

Describe the correlation between age and BMI.

[1]

(c) Use the line of best fit to estimate the BMI of a 30-year-old man.

[1]

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(d) Explain why it would not be sensible to use the line of best fit to estimate the BMI of a 60-year-old man.

[1]

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- (e) Use your knowledge of the large data set to suggest two reasons why the sample data in the table may not be representative of the population.

[2]

- (f) Once the data in the large data set had been cleaned there were 196 values available for selection. Describe how a sample of size 12 could be generated using systematic sampling so that each of the 196 values could be selected in the sample.

[2]

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49(a) Fig. 4 shows a cumulative frequency diagram for the time spent revising mathematics by year 11 students at a certain school during a week in the summer term.

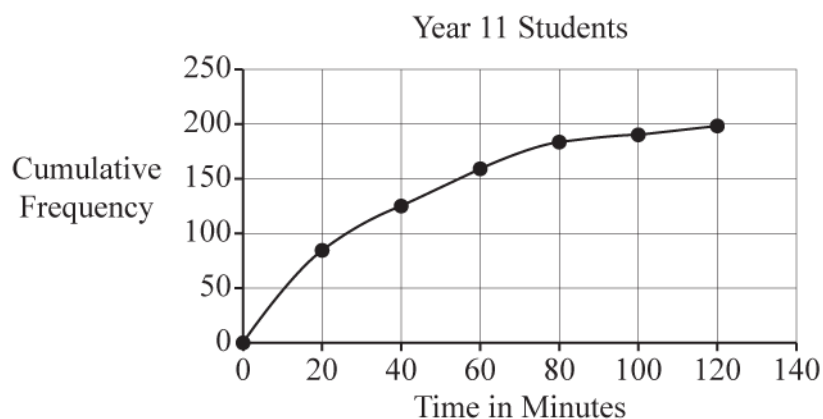


Fig. 4

Use the diagram to estimate the median time spent revising mathematics by these students.

[1]

(b) A teacher comments that 90% of the students spent less than an hour revising mathematics during this week.

Determine whether the information in the diagram supports this comment.

[1]

50(a) Rosella is carrying out an investigation into the age at which adults retire from work in the city where she lives. She collects a sample of size 50, ensuring this comprises of 25 randomly selected retired men and 25 randomly selected retired women.

State the name of the sampling method she uses.

[1]



(b) Fig. 8.1 shows the data she obtains in a frequency table and Fig. 8.2 shows these data displayed in a histogram.

Age in years at retirement	45 –	50 –	55 –	60 –	65 –	70 –	75 – 80
Frequency Density	0.4	1.8	2.4	2.2	1.8	1.2	0.2

Fig. 8.1

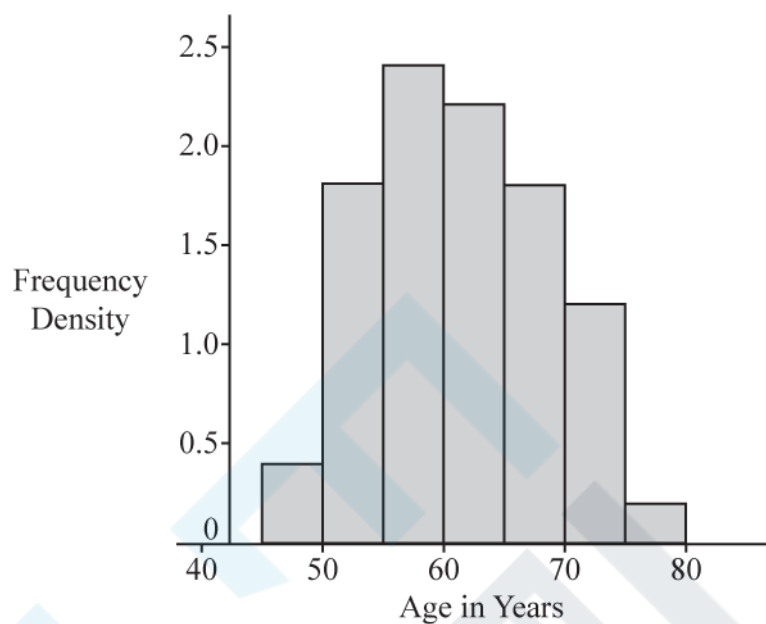


Fig. 8.2

How many people in the sample are aged between 50 and 55?

[1]

- (c) Rosella obtains a list of the names of all 4960 people who have retired in the city during the previous month.

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Describe how Rosella could collect a sample of size 200 from her list using

- systematic sampling such that every item on the list could be selected,
- simple random sampling.

[4]



- (d) Rosella collects two simple random samples, one of size 200 and one of size 500, from her list. The histograms in Fig. 8.3 show the data from the sample of size 200 on the left and the data from the sample of 500 on the right.

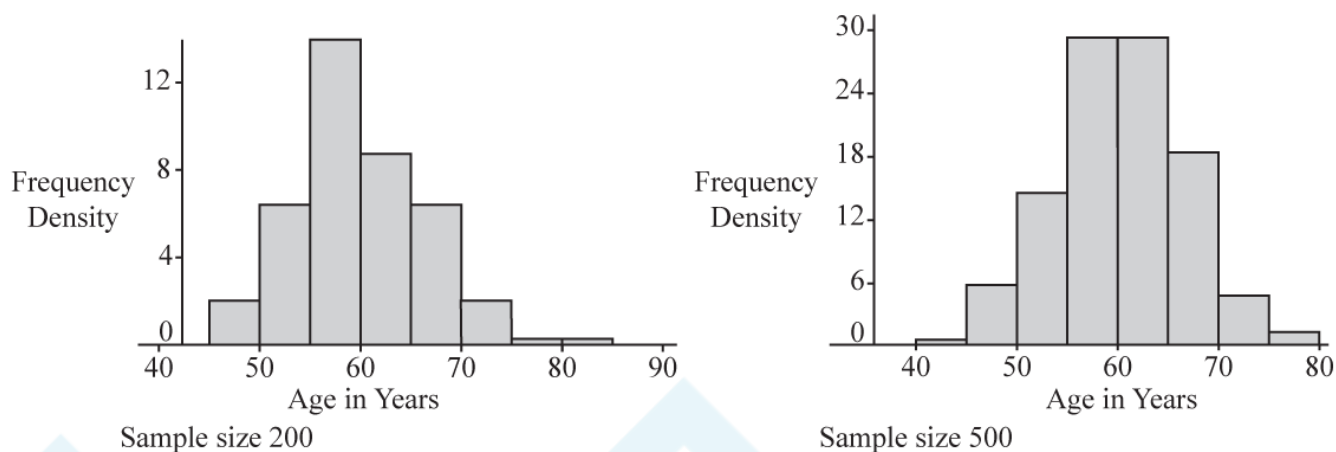


Fig. 8.3

With reference to the histograms shown in Fig. 8.2 and Fig. 8.3, explain why it appears reasonable to model the age of retirement in this city using the Normal distribution. [1]

- (e) Summary statistics for the sample of 500 are shown in Fig. 8.4.

Statistics	
n	500
Mean	60.0515
σ	6.5717
s	6.5783
Σx	30025.7601
Σx^2	1824686.322
Min	36.0793
Q1	55.2573
Median	59.9202
Q3	64.4239
Max	81.742

Fig. 8.4

Use an appropriate Normal model based on the information in Fig. 8.4 to estimate the number of people aged over 65 who retired in the city in the previous month.

[4]

- (f) Identify a limitation in using this model to predict the number of people aged over 65 retiring in the following month.

[1]

51(a) The pre-release material contains information concerning median house prices over the period 2004 – 2015. A spreadsheet has been used to generate a time series graph for two areas: the London borough of “Barking and Dagenham” and “North West”. This is shown together with the raw data in Fig. 11.1.

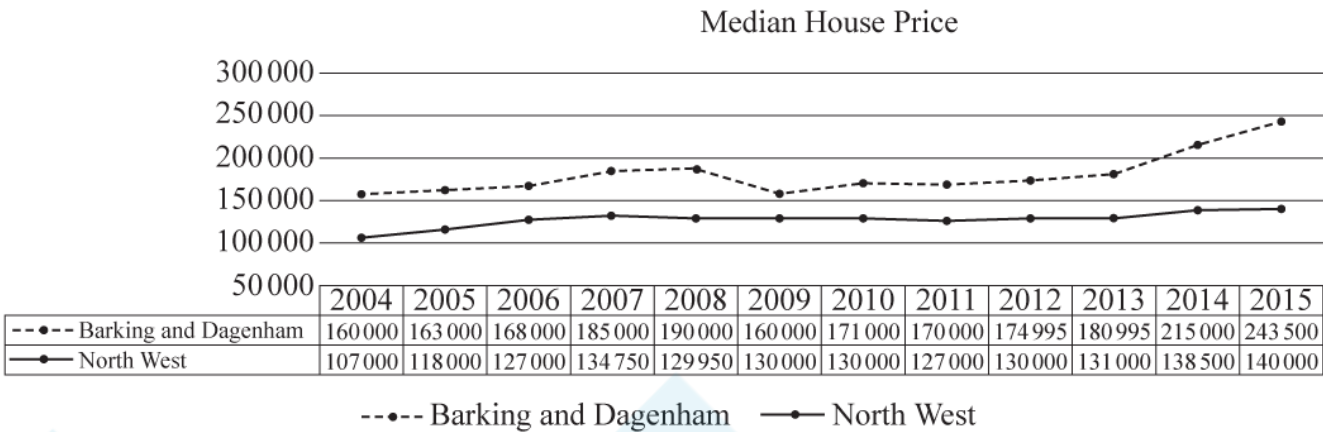


Fig. 11.1

Dr Procter suggests that it is unusual for median house prices in a London borough to be consistently higher than those in other parts of the country.

Use your knowledge of the large data set to comment on Dr Procter’s suggestion.

[1]

- (b) Dr Procter wishes to predict the median house price in Barking and Dagenham in 2016. She uses the spreadsheet function LINEST to find the equation of the line of best fit for the given data. She obtains the equation

$P = 4897Y - 9\,657\,847$, where P is the median house price in pounds and Y is the calendar year, for example 2015.

Use Dr Procter's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]

- (c) Professor Jackson uses a simpler model by using the data from 2014 and 2015 only to form a straight-line model.

Find the equation Professor Jackson uses in her model.

[2]

- (d) Use Professor Jackson's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]

- (e) Professor Jackson carries out some research online. She finds some information about median house prices in Barking and Dagenham, which is shown in Fig. 11.2.

2016	2017
£290 000	£300 000

Fig. 11.2

Comment on how well

- Dr Procter's model fits the data,
- Professor Jackson's model fits the data.

[2]

- (f) Explain which, if any, of the models is likely to be more reliable for predicting median house prices in Barking and Dagenham in 2020.

[1]

52(a) In this question you must show detailed reasoning.

A 5-sided spinner can give scores of 1, 2, 3, 4 or 5. After observing a large number of spins, Elaine models the probability distribution of X , the score on the spinner, as shown in Fig. 12.

x	1	2	3	4	5
$P(X = x)$	0.2	0.3	p	p	q

Fig. 12

When the spinner is spun twice, the probability of obtaining a total score of 9 is 0.06.

Given that $q < 2p$, determine the values of p and q .

[6]

(b) The spinner is spun 10 times. Calculate the probability that exactly one 5 is obtained.

[2]

- (c) Elaine's teacher believes that the probability that the spinner shows a 1 is greater than 0.2. The spinner is spun 100 times and gives a score of 1 on 28 occasions.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to suggest that the probability of obtaining a score of 1 is greater than 0.2. [7]

53(a) The pre-release material contains information concerning median house prices, recycling rates and employment rates. Fig. 13.1 shows a scatter diagram of recycling rate against employment rate for a random sample of 33 regions.

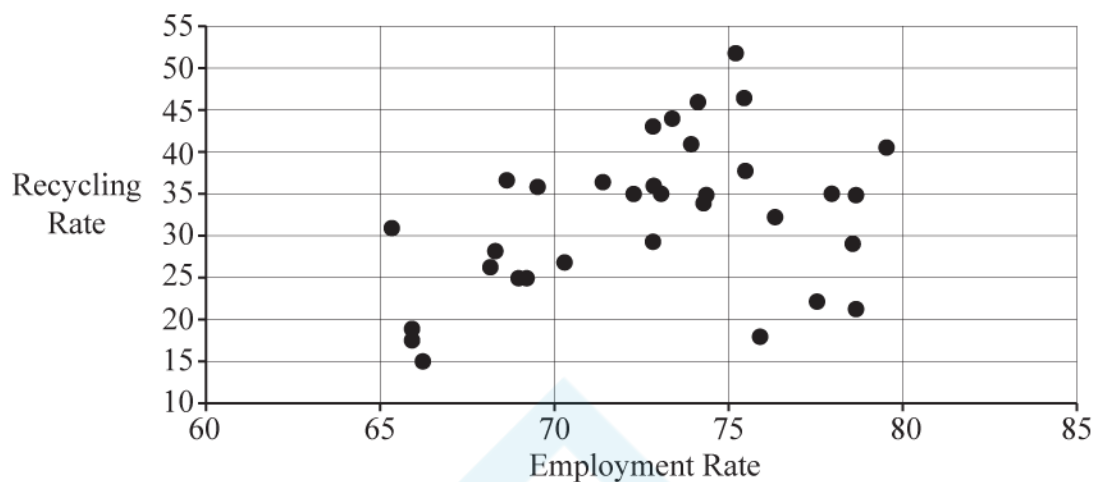


Fig. 13.1

The product moment correlation coefficient for this sample is 0.37154 and the associated p -value is 0.033.

Lee conducts a hypothesis test at the 5% level to test whether there is any evidence to suggest there is positive correlation between recycling rate and employment rate. He concludes that there is no evidence to suggest positive correlation because $0.033 \approx 0$ and $0.37154 > 0.05$.

Explain whether Lee's reasoning is correct.

[2]

- (b) Fig. 13.2 shows a scatter diagram of recycling rate against median house price for a random sample of 33 regions.

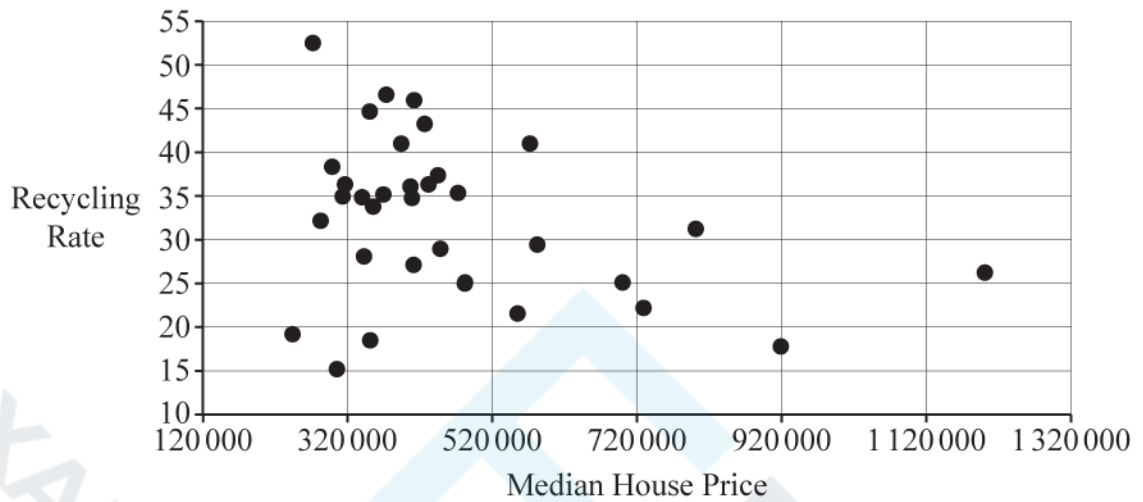


Fig. 13.2

The product moment correlation coefficient for this sample is -0.33278 and the associated p -value is 0.058 .

Fig. 13.3 shows summary statistics for the median house prices for the data in this sample.

Statistics	
n	33
Mean	465467.9697
σ	201236.1345
s	204356.2606
Σx	15360443
Σx^2	8486161617387
Min	243500
Q1	342500
Median	410000
Q3	521000
Max	1200000

Fig. 13.3

Use the information in Fig. 13.3 and Fig. 13.2 to show that there are at least two outliers.

[2]

(c) Describe the effect of removing the outliers on

- the product moment correlation coefficient between recycling rate and median house price,
- the p -value associated with this correlation coefficient, in each case explaining your answer.

[2]

(d) All 33 items in the sample are areas in London. A student suggests that it is very unlikely that only areas in London would be selected in a random sample.

Use your knowledge of the pre-release material to explain whether you think the student's suggestion is reasonable.

[1]

END OF QUESTION PAPER